

Article

A Systemic Approach to Sustainable Technological Extension: A Dynamic Model for Oil Palm Cultivation in Colombia

Julián F. Becerra-Encinales ^{1,2,*} , Jair D. Rodríguez-Ortega ², Alexandre P. Cooman ¹, Luis H. Reyes ² 
and Juan C. Cruz ² 

¹ Colombian Oil Palm Research Center Corporation—Cenipalma, Bogotá 111121, Colombia

² School of Engineering, Los Andes University, Bogotá 111711, Colombia;
jd.rodriguez01@uniandes.edu.co (J.D.R.-O.); lh.reyes@uniandes.edu.co (L.H.R.);
jc.cruz@uniandes.edu.co (J.C.C.)

* Correspondence: jbecerra@cenipalma.org

Abstract: This study develops a systemic conceptual model to analyze the dynamics of technological extension and its impact on the sustainability of Colombia's oil palm production sector. Using a system dynamics-based methodological approach, multiple sources of information were integrated, including the Sustainability Index (SI), socio-economic characterizations of producers, and ethnographic insights, with the aim of identifying key actors and their interactions within the system, modeling causal relationships and feedback loops. In contrast with the traditional linear model of technology transfer, this model reveals non-linear dynamic structures, highlighting ten systemic loops (six reinforcing and four balancing) that explain how technological adoption, legal and regulatory compliance, and the SI risk level interact through dynamic feedback mechanisms. Beyond providing a structural representation of the system, the model offers concrete guidance for policy formulation and intervention strategies tailored to different producer typologies. Theoretically, this research broadens the understanding of diffusion, extension, and technology adoption processes in complex agricultural systems in developing countries. Practically, it provides a tool for analysis and planning to strengthen the sustainability of the Colombian oil palm sector, with potential applicability to other agro-industrial contexts. Future research should advance toward dynamic simulations to evaluate intervention scenarios, emphasizing the need for progressive incentive schemes, participatory extension, and integrated regulatory frameworks that promote high levels of long-term sustainability compliance.

Keywords: system dynamics; sustainable agriculture systems; agricultural extension model; technological adoption; oil palm



Academic Editor: Spyros Fountas

Received: 24 March 2025

Revised: 5 May 2025

Accepted: 16 May 2025

Published: 20 May 2025

Citation: Becerra-Encinales, J.F.; Rodríguez-Ortega, J.D.; Cooman, A.P.; Reyes, L.H.; Cruz, J.C. A Systemic Approach to Sustainable Technological Extension: A Dynamic Model for Oil Palm Cultivation in Colombia. *Sustainability* **2025**, *17*, 4706. <https://doi.org/10.3390/su17104706>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Systemic models in agriculture have been globally implemented as essential tools for management and decision making. In Australia, for example, these models have facilitated the sustainable management of water resources in agriculture [1], while, in Brazil, they have been crucial for optimizing fertilizer use in soybean cultivation and minimizing environmental impact [1,2]. In developing countries, systemic models are also being applied, as seen in Indonesia, where they have been implemented to increase rice production, a critical component of the country's food security. This approach has helped researchers and policymakers gain a clear understanding and manage the interaction between various factors, such as rice-growing areas, productivity, and post-harvest practices, which is essential for improving rice production quantity and quality [3].

Similarly, in Colombia, system dynamics have been used to address food sufficiency, particularly in managing land use and food demand [4]. These models have demonstrated that increasing productivity and efficiency is crucial beyond simply expanding agricultural land. This approach has helped establish sustainability-driven behaviors among producers, service infrastructure adjustments, and irrigation and drainage system improvements [4]. These examples highlight how systemic models can be applied across different agricultural sectors to enhance efficiency and sustainability.

Despite the widespread use of systemic approaches, their application in agriculture remains limited, particularly in technological extension. Emerging economies face unique challenges that could significantly benefit from systemic models, such as understanding and strategically defining technological extension [4,5], and facilitating the adoption of advanced technologies and sustainable practices [5]. The challenges include producer resistance to change, adverse political and economic contexts, high variability in cultivation conditions, and region-specific contexts. In contrast, the opportunities lie in enhancing productivity through technological innovation that fosters economic, environmental, and social sustainability [3–8]. In Colombia, and more specifically in oil palm cultivation, integrating systemic models that lead to efficiency and optimization in technology transfer and adoption is practically non-existent. Such an absence represents a significant gap in the literature and theoretical knowledge regarding developing technological extension strategies for this productive sector in the Colombian context.

The current literature on technology transfer and adoption in this sector has primarily focused on traditional extension approaches based on the linear transfer of knowledge, where innovation dissemination originates from external actors without a systemic integration with producers and other stakeholders involved [9–11]. This perspective has limited the adaptability of extension models to the heterogeneous reality of the Colombian oil palm sector, which is characterized by a diverse range of producers with varying levels of access to technology, infrastructure, financing, and other critical factors influencing adoption [12,13].

In the Colombian oil palm sector, the diffusion of technology faces significant structural and institutional barriers that disproportionately affect smallholder producers. These include unequal access to credit, technical assistance, and infrastructure, contrasting sharply with the advantages enjoyed by large-scale producers [13]. Furthermore, there are critical asymmetries in access to updated information, decision-making tools, and connectivity, which constrain the capacity of smallholders to adopt sustainable innovations. The diffusion system is also characterized by fragmentation among institutions, limited coordination among extension agents, and inconsistencies in the delivery of technical messages. These structural dilemmas create inertia and reduce the effectiveness of traditional extension strategies, underscoring the urgency to develop more systemic and inclusive approaches to technological extension [13,14].

Our study examines how a system dynamics-based modeling tool enhances our understanding of the relationship between extension and technological adoption by addressing the following research question: how can a systemic model for technological extension improve the adoption of practices, technologies, and innovations with economic, environmental, and social sustainability in oil palm cultivation in Colombia? The need for a systemic approach becomes evident when considering the complexity of variables involved in this sector, which requires a model that reflects the current reality and projects the future dynamics of a changing environment with diverse actors and influencing factors. Our research is based on the methodology of Sterman (2000) [15], which involves developing a model that incorporates system dynamics to provide a precise and functional representation of technological adoption processes. Traditional models, often linear and non-systemic,

have failed to capture the interdependence and feedback among economic, environmental, and social variables that characterize contemporary agriculture [5,16,17].

Based on our experience, learning-based models, such as the one used in our research, are helpful because they explicitly highlight the factors and relationships that influence or limit adoption decisions by producers, different actors, or involved institutions. These models transform the mental adoption model into a complex, multifactorial, and multidimensional actual model. As a contribution to knowledge, our study formulated a dynamic hypothesis identifying two additional key variables alongside (a) technological adoption, (b) legal and regulatory compliance, and (c) SI (sustainability index) risk level. These three interrelated variables hold equal importance and demand attention. This contribution provides a solid foundation for future research in system dynamics, leading to the development of simulation models, model validation, and policy interventions to achieve sustainable oil palm production in Colombia. Additionally, the conceptual basis of this research can be applied to other productive sectors and developing countries.

The remainder of this paper is structured as follows. Section 2 describes the methodological framework, detailing the application of system dynamics and the triangulation of multiple data sources. Section 3 presents the main findings and discusses the structure of the conceptual model, including the identification of key actors, decision rules, and the feedback loops derived from the three core variables of the dynamic hypothesis: technological adoption, legal and regulatory compliance, and SI risk level. Section 4 provides the main conclusions of this study and proposes future research directions based on the theoretical and practical implications of the model.

2. Methodology

2.1. Context and Location

This study was conducted within the context of oil palm cultivation in Colombia, a sector that, in 2023, accounted for 17.6% of the national agricultural GDP [18,19]. With approximately 580,000 hectares under cultivation, production is distributed across four oil palm zones, each with distinct geographic and climatic characteristics that influence the sustainability of the crop. Colombia is the world's fourth-largest palm oil producer and the leading producer in the Americas, with a presence in 155 municipalities across 20 departments in the country [19].

2.2. Research Design

This research employed a mixed methods approach, integrating both quantitative and qualitative methodologies to ensure a systemic and robust analysis [4,15,20]. This combination allowed for a comprehensive understanding of the problem by using historical data and expert knowledge to identify mental models and construct an actual model that explains technological extension in the Colombian oil palm sector [4].

To support the development of the model and the formulation of the dynamic hypothesis, this study triangulated three complementary data sources: (a) the sustainability index (SI), which provided a classification of producers based on multidimensional sustainability performance; (b) quantitative socioeconomic profiles, which characterized variables such as age, education, and production scale; and (c) qualitative data from ethnographic interviews. These sources were cross-analyzed iteratively to identify patterns, validate relationships, and guide the structuring of causal assumptions and actor dynamics. This triangulation ensured a robust foundation for model construction by integrating both measurable indicators and grounded perspectives from the field.

Interview transcripts were analyzed using thematic coding techniques. The coding process focused on identifying variables related to adoption behavior, perceived barriers,

ers, actor interactions, and contextual dynamics. Codes were grouped under theoretical categories aligned with system dynamics concepts, such as delays, feedback, goals, and resources. Cross-validation with SI classification and socioeconomic data allowed for the convergence and complementarity of findings, strengthening the internal consistency of the model structure and the plausibility of its feedback loops.

Interviewees were selected using a purposive sampling strategy based not only on their SI classification but also on additional dimensions such as geographic location, farm size, level of market articulation, and organizational affiliation (e.g., independent producers vs. those associated with producer groups or mills). This ensured that the model captured a diverse range of experiences and perspectives within the oil palm sector.

The ethnographic interviews were guided by a semi-structured protocol organized around three core themes: (1) producers' perceptions of technological extension and innovation; (2) perceived barriers and facilitators for adoption; and (3) interactions with external actors and institutions. These themes are further detailed in Appendix A (Table A1). Interviews continued until thematic saturation was reached, that is, when no new conceptual categories emerged from subsequent transcripts. To identify actor roles, interests, and causal assumptions, we analyzed patterns in how interviewees described their decision-making processes, perceived responsibilities, and expectations regarding outcomes. These insights were essential for accurately assigning behavioral attributes and feedback influences within the model.

- Quantitative approach: Data collection and analysis focused on key indicators such as producer typologies based on the sustainability index (SI), socioeconomic characterization, and ethnographic assessments. These datasets facilitated the identification of patterns and trends within the sector's behavior.
- Qualitative approach: Generative sessions were conducted with focus groups and key system actors to determine their roles, interests, and the causal relationships influencing their decision-making processes. Additionally, the conceptual model was validated through iterative consultations with experts.

This mixed methods design enabled data triangulation, ensuring consistency between quantitative and qualitative findings. Furthermore, it provided a foundation for dynamic system modeling, accurately representing the sector's reality. This approach successfully transformed a mental model into an actual model for technological extension in Colombia's sector of oil palm cultivation (Figure 1).

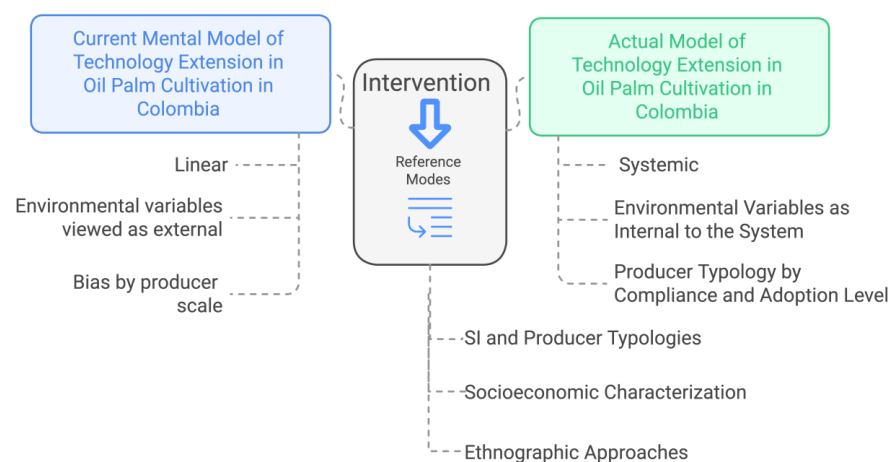


Figure 1. Reference modes used for the transformation of the mental model. The current mental model is represented in blue, while the actual model derived from intervention through analysis and appropriation of the reference modes is shown in green.

2.3. Methodological Approach

The methodological approach is based on the five-step iterative model proposed by Sterman (2000) [15], which is widely used in system dynamics studies for analyzing complex problems. The five steps of the process are shown in Table 1 [4,15].

Table 1. Iterative model phases based on Sterman (2000) [15].

Steps	Description
Problem articulation	Clearly defining the issue of technology extension in the Colombian oil palm sector, establishing its reference modes, system boundaries, and time horizon.
Dynamic hypothesis	Identifying key actors and their interactions, constructing causal relationships and feedback loops, and determining the system's dynamics.
Formulation of the simulation model	Developing a model based on differential equations and decision rules that represent the change in variables over time.
Validation tests	Evaluating the robustness and coherence of the model to ensure its predictive capability and representativeness of the actual system.
Policy design and evaluation	Formulating strategies to improve technology extension, considering structural leverage points within the system.

Although the methodological approach encompasses these five steps, this study focuses on the first two—problem articulation and the dynamic hypothesis—establishing the foundation for the system's dynamic analysis (Figure 2). Previous studies in system dynamics have demonstrated that problem articulation, which integrates the identification of key actors and the formulation of the dynamic hypothesis, represents the most complex steps while providing a solid analytical framework for developing dynamic systems.

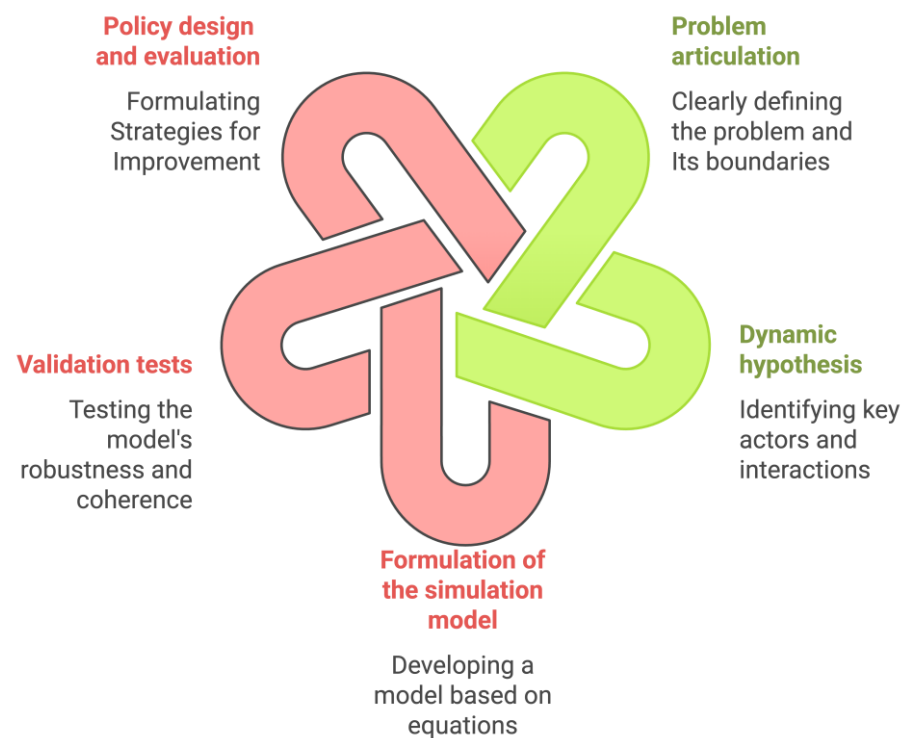


Figure 2. Iterative model based on Sterman (2000) [15,20]. The steps of the iterative model developed in this study are represented in green, while those to be addressed in future research are shown in pink.

2.3.1. Problem Articulation

The first step involved defining the problem [4,15]. In this study, the management of technological extension in the oil palm sector in Colombia was identified as a critical issue due to existing gaps in technological adoption and legal and regulatory compliance, which limit the sector's competitiveness and sustainability.

To articulate the problem, the following key aspects were considered:

- a. Reference modes: sustainability index (classification of producers based on the level of adoption of practices, technologies, and innovations, and legal and regulatory compliance) [14], socioeconomic characterization of producers [13], and ethnographic approaches through semi-structured interviews using a questionnaire described in Appendix A (Table A1).
- b. Time horizon: The system's behavior was analyzed over a five-year horizon, considering both baseline information from the reference modes and the historical causes of the problem, along with the potential future implications of its resolution. Future periodic data collection from the reference modes will provide the analytical foundation for formulating the simulation model, validation, and designing and evaluating intervention policies.
- c. Model boundary: Factors related to technological extension were defined, excluding those that did not directly influence the identified problem. Consequently, research and development (R&D) processes and interactions were excluded. This decision was based on the fact that, although R&D activities are highly important, they largely occurred outside the decision-making cycles of producers, extension agents, and other related actors, and were not directly influenced by the feedback loops modeled in this study. As shown in Figure 3, the model boundary extended from extension as an output variable to adoption as an impact or response variable. Adoption, according to the model boundary, included both the application of technology and compliance with laws and regulations in Colombia's oil palm industry. Acknowledging this boundary, one limitation was that potential innovations or changes in available technologies originating from external research were treated as exogenous inputs rather than being endogenized within the model.

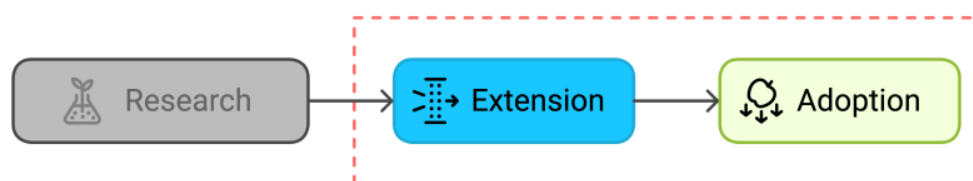


Figure 3. Model boundary for the dynamic hypothesis. This boundary integrates extension as an output variable and adoption as an impact variable. The dashed box represents the internal scope of the model, which excludes research activities and focuses on the feedback dynamics between extension and adoption.

Based on the research question (how can a systemic model for technological extension improve the adoption of practices, technologies, and innovations with economic, environmental, and social sustainability in oil palm cultivation in Colombia), we determined the dynamics of technological extension and identified key actors and their roles based on the following categories proposed by Olaya (2016) [20]:

- a. Executors: who carries out actions that directly impact the problematic situation?
- b. Collaborators: who performs actions or provides resources or information that facilitate the actions of the executors?
- c. Beneficiaries: who benefits from the problematic situation?

- d. Affected parties: who suffers adverse effects from the problematic situation?
- e. Responsible parties: who is accountable for resolving or improving the problematic situation?
- f. Interveners: who, as part of the environment or context, can provide opportunities or pose threats that may improve or worsen the problematic situation?

Each actor has specific interests and goals that influence technological extension, establishing causal relationships within the system.

2.3.2. Dynamic Hypothesis

At this stage, a conceptual model was developed based on a dynamic hypothesis explaining how feedback structures within the system generate the observed behavior (Figure 4). This includes the characterization of decision rules, where each actor makes decisions based on specific criteria. The identification and definition of decision rules for each actor (inputs, decision variables, and outputs) were guided by a thematic analysis of the ethnographic interview transcripts. Recurring patterns in how actors described their decision-making processes were coded and grouped by actor type (e.g., producers, extensionists, regulators), and then cross-validated using data from the sustainability index and socioeconomic profiles. For instance, technical information frequently cited by multiple producers as a basis for action was coded as an “input”, whereas judgments involving cost-benefit trade-offs or regulatory pressure were coded as “decision variables”. Finally, actions commonly mentioned—such as investing in technology or attending training—were coded as “outputs”. This triangulated process ensured that the decision rules reflected both empirical insights and theoretical alignment with prior agricultural systems research. To characterize decision rules, the following elements were identified [15,20,21]:

- a. Inputs: technical information, economic incentives, regulatory frameworks.
- b. Decision variables: assessment of costs, benefits, and regulatory constraints.
- c. Outputs: actions such as investment in technology, regulatory compliance, and producer training.

This analysis enabled a comprehensive understanding of how actors interact and shape the system’s dynamics. The key elements include the following:

- a. Causal relationships and polarity: connections between actors and variables were established, defining whether a relationship is reinforcing (+) or balancing (−).
- b. Key variables: technological adoption, legal and regulatory compliance, and the SI risk level.
- c. Feedback loops: reinforcing loops that accelerate technological adoption and legal compliance were identified, along with balancing loops that regulate the SI risk level.

The system dynamics-based methodological approach provided an analytical framework for understanding technological extension in the oil palm sector as a complex social system where all actors make decisions simultaneously. Future research should focus on developing simulation models, validating results, and designing intervention policies to enhance sustainability in oil palm cultivation. These efforts align with steps 3, 4, and 5 of the iterative model proposed by Sterman (2000) [1–3,5,15,20,21].

The causal loop diagrams presented in this study were constructed using Vensim® PLE 10.2.2 (Ventana Systems, Inc., Harvard, MA, USA), a specialized software for system dynamics modeling. This tool allowed for the visualization of reinforcing and balancing feedback loops and the representation of actor interactions across sustainability dimensions.

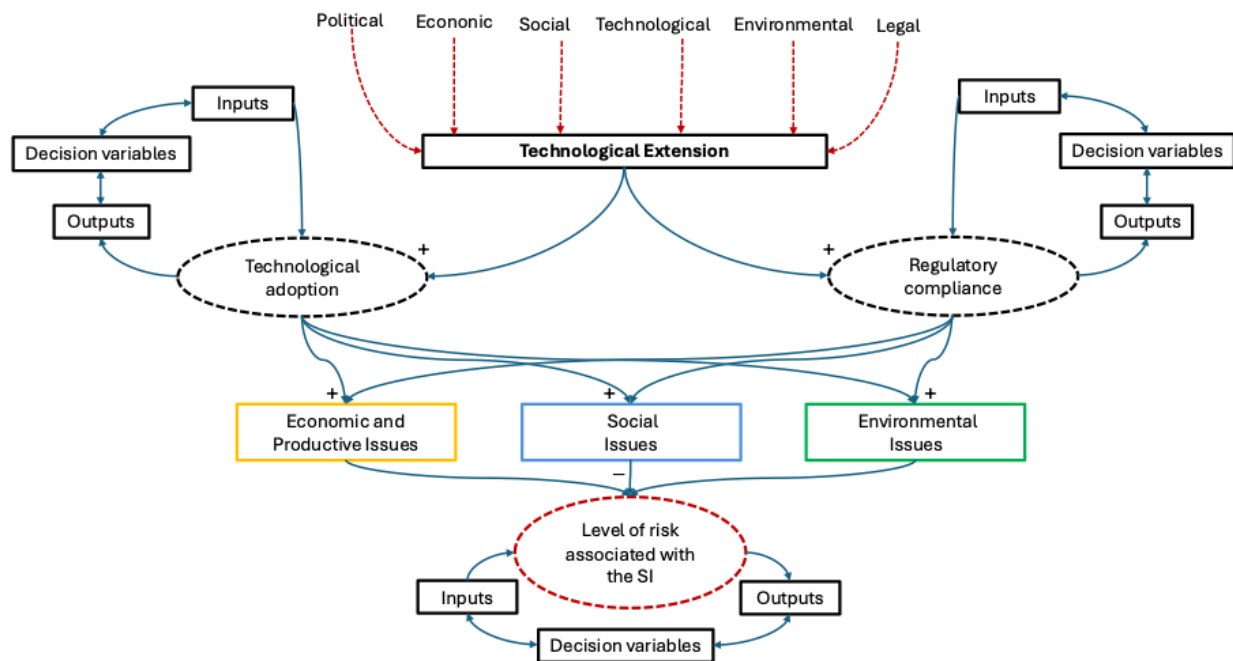


Figure 4. Diagram for the characterization of decision rules. This perspective integrates sustainability's economic, environmental, and social dimensions and external pressures from political, economic, social, technological, environmental, and legal (PESTEL) variables [22] within the specific environments and contexts of social knowledge appropriation. Solid arrows represent direct causal relationships, while dashed arrows indicate external influences or contextual pressures.

3. Results and Discussion

3.1. Problem Articulation Based on the Analysis of Reference Modes and Their Context in the Existing Literature

The analysis of reference modes enabled the identification of the evolution and structure of technological extension in the Colombian oil palm sector. This study relied on three primary sources of information: (1) the sustainability index (SI) [14], (2) socioeconomic characterization [13], and (3) ethnographic approaches. Structural patterns explaining the dynamics of technological adoption and regulatory compliance among oil palm producers were identified. These findings were compared with previous studies on the dynamics and adoption patterns of agricultural technologies, allowing an evaluation of how this research contributes to understanding the factors driving or limiting sustainability in agro-industrial production.

3.1.1. Analysis of the Sustainability Index (SI) and Its Correlation with Technological Adoption

The SI has been established as a key analytical tool for assessing and monitoring sustainability in oil palm cultivation in Colombia. This metric measures the adoption level of 79 agricultural practices, organized into 29 thematic categories, aligned with the 10 principles and 3 fundamental pillars of sustainability: economic, environmental, and social (as defined for oil palm cultivation in Colombia, as described in Appendix B (Table A2)) [18,19,23]. Additionally, the SI incorporates regulatory compliance and productive performance variables, providing a comprehensive overview of sustainability status within units for oil palm production.

The validity and reliability of the SI have been evaluated through internal validation processes and consultations with experts in the agro-industrial sector, ensuring that the criteria used in its construction accurately reflect the productive and regulatory conditions of the country [17]. Previous studies have documented the effectiveness of the SI as an

instrument for differentiating technology adoption levels and regulatory compliance among producers, facilitating the design of intervention and technological extension strategies [14]. However, given that the SI is an evolving tool, continued sensitivity analyses and statistical reliability studies are recommended to further strengthen its applicability and precision in decision making within the Colombian oil palm sector.

For this study, a total of 3808 SI records were analyzed, enabling the classification of producers based on their level of technological adoption and regulatory compliance within the framework of sustainability [24]. The data collected and analyzed revealed the existence of three distinct groups, as shown in Figure 5.

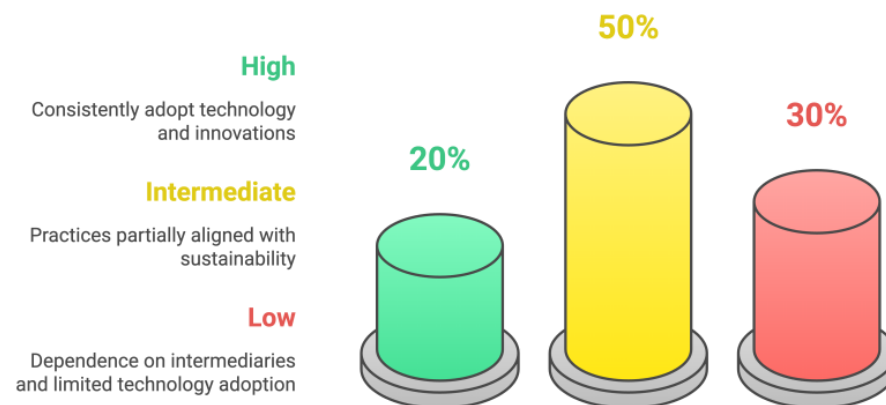


Figure 5. Distribution of producers by sustainability level. This study conducted a global analysis of 3808 sustainability impact assessments in the context of Colombian oil palm cultivation. Green represents the percentage of SI assessments with high compliance levels, yellow indicates the percentage of SI assessments with intermediate compliance levels, and red denotes the percentage of SI assessments with low compliance levels.

Producers with high sustainability: This category is characterized by the continuous adoption of technological innovations, a high degree of legal and regulatory compliance, and access to conducive environments that support a high level of SI compliance. They represent 20% of the total SI records and concentrate the highest productivity within the sector.

Producers with intermediate sustainability: These producers implement agricultural practices that are partially aligned with sustainability standards and legal compliance but face limitations in their environment, such as restricted access to technical assistance, financing, and other key variables. They constitute approximately 50% of all producers.

Producers with low sustainability: this segment (30% of producers) exhibits low technological adoption, difficulties in legal and regulatory compliance, and likely a strong dependency on intermediaries for market access and financing.

The results of this analysis align with previous findings in agricultural production systems, where access to technical assistance within extension models that facilitate knowledge transfer, financial resources, credibility, access to technologies, and availability of skilled labor are key determinants of technological adoption [9,11,12,25,26]. However, this study did not capture the transition between sustainability levels, a process that tends to be slow according to research on structural and socioeconomic barriers limiting the adoption of new technologies in perennial crops [10,27], such as oil palm.

3.1.2. Socioeconomic Characterization and Its Impact on Technological Extension

The socioeconomic characterization process of oil palm producers was completed in 2024, gathering data from 7665 producers of different scales at the national level. This process has enabled the consolidation of information on the livelihoods of oil palm growers,

providing a baseline for designing technical assistance strategies tailored to the social profiles of producers [13].

The socioeconomic characterization of oil palm producers in Colombia was conducted through a structured digital questionnaire divided into modules addressing key aspects such as general information, housing conditions, income, social organization, and access to technical assistance. The collected data facilitated an understanding of producers' demographic and cultural diversity, which will aid in the future development of differentiated and intersectional extension strategies.

The socioeconomic characterization revealed significant differences among producers based on education level, access to financing, and organizational models [13]. The results indicate that, by the end of 2024, most producers belonged to the small-scale (≤ 20 ha) or medium-scale (>20 ha <500 ha) category, with geographic distribution across 143 municipalities in 20 Colombian departments. Technological extension has been crucial in this characterization, as most producers have been characterized through technical assistance units managed by extraction companies, followed by independent producers characterized by the Oil Palm Research Center (Cenipalma).

Another key aspect was the cultural identification of producers. The characterization indicates that 42% of producers identified as peasants, 29% as Afro-descendants, and 28% did not identify with any specific category. This evidence suggests that cultural identity varies significantly depending on production scale and geographic location.

Education level was also a determining factor in technological adoption. Eighty percent of small-scale producers did not have higher education, whereas, among large-scale producers, this figure was only twenty percent. Additionally, 6.4% of producers reported being unable to read or write, highlighting the need to adapt extension technology programs and models to different educational levels.

Regarding living conditions, multidimensional poverty affected 19% of the palm-growing population, with critical dimensions such as low educational attainment (65.2%) and informal employment (61.6%). This situation underscores the importance of technical training and social assistance programs within technological extension strategies.

Another relevant aspect was the residency pattern of producers. A total of 70.4% of palm growers did not reside on the farm where their crop was located, suggesting that technological extension models should consider the role of farm managers and workers in decision-making processes.

Previous studies have demonstrated that education level and economic resources significantly influence producers' willingness to adopt innovative practices [28–30]. In this regard, the findings of our research reinforce the idea that producers with lower education levels and limited access to financing face more significant barriers to implementing new technologies [31–33]. Additionally, it was found that producers who were part of associations or cooperatives had a higher rate of technological adoption due to the presence of support networks that facilitate knowledge transfer, which has been widely documented in studies on participatory agricultural extension [28,34–37].

3.1.3. Ethnographic Approaches and Mental Models on Technological Extension

The sample for ethnographic approaches was determined based on producer typologies proposed by Becerra-Encinales (2025) [14]. A total of 50 semi-structured interviews were conducted with producers for each of the established typologies, ensuring an equitable distribution among different levels of technological adoption and regulatory compliance within the SI. The selection of producers was based on the following specific criteria:

- Level of SI compliance: producers were prioritized based on their classification within the three sustainability levels (high, intermediate, and low) for each axis (economic, environmental, and social), according to SI results.
- Geographical distribution: producers from the four main oil palm-growing regions in Colombia (Northern, Central, Eastern, and Southwestern) were included to capture the heterogeneity of agroecological and socioeconomic conditions.
- Organizational model: independent producers, those associated with cooperatives, and those linked to oil palm mills were considered in evaluating differences in technology adoption based on production structures.
- Production scale: producers of different farm sizes (small, medium, and large-scale) were included to analyze the influence of production scale on technological extension.
- Access to technical assistance: producers with and without regular access to extension programs were selected to compare the impact of technical support on decision making.
- Availability and consent: only producers who voluntarily agreed to participate in the study were interviewed, ensuring compliance with ethical and methodological research principles.

Additionally, participatory observations were carried out in production units, allowing for the validation of perceptions expressed in the interviews and a more profound understanding of the existence of divergent mental models among producers. These findings provided key insights for identifying actors and key variables in constructing a real model under a systemic approach to technological extension in the oil palm sector.

Key findings include the following:

- I. Factors influencing technological adoption: When producers were asked about the factors that motivate or influence their decision to implement practices, technologies, and innovations in their production process, 78% agreed that economic factors are decisive. In comparison, 43.14% attributed this phenomenon to labor availability (Figure 6).

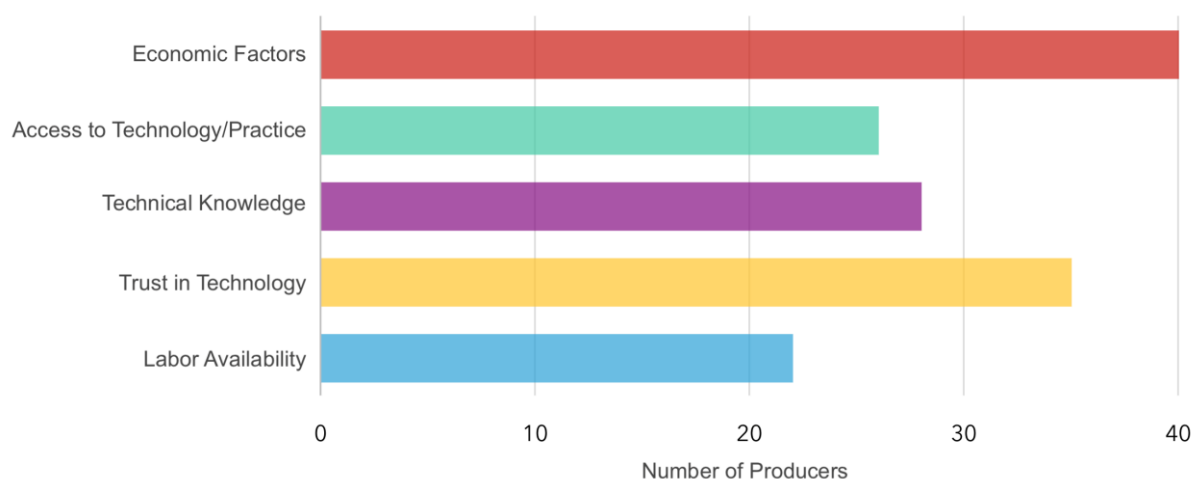


Figure 6. Influencing or motivating factors for implementing sustainability-oriented practices, technologies, and innovations identified through ethnographic approaches with oil palm producers in Colombia.

Figure 6 illustrates that economic factors were the primary determinant in technological adoption decisions, significantly surpassing other evaluated factors. The evidence suggests that investment costs, expected profitability, and access to financing play a central role in producers' decision making. These findings align with previous studies that

identify access to financial resources as a key determinant in agricultural technology adoption [9,29,38].

Another significant finding in our study was trust in technology, which ranked second in importance. This study indicates that the perceived efficacy and utility of a practice, a technology, or an innovation greatly influence producers' willingness to implement it. The literature on technology adoption highlights that producers are often reluctant to incorporate new practices when they are unaware of their benefits or have no prior experience with them [10,26,39].

Technical knowledge also emerged as a relevant factor, demonstrating that training and education influence the adoption of new technologies. Such evidence underscores the importance of extension programs and technical assistance in promoting innovations within the oil palm sector. Previous studies have shown that producers with access to technical information and specialized training exhibit higher technological adoption rates and legal and regulatory compliance [28,39].

We also identified access to technology/practices and labor availability as influencing factors, although they carried a lower relative weight. However, based on our experience, these factors will become increasingly important, as, in extensive agricultural systems, the availability and qualification of workers can represent a significant barrier to the adoption of technological innovations [38,40–44].

Furthermore, other observed factors that align with the literature include the following:

Perception of technology: while large-scale producers consider technology a strategic investment, small-scale producers perceive it as an additional cost without immediate benefits [45,46].

Role of extension agents: differences were identified in the perception of the extensionist's role; some producers see them as facilitators of technological change, while others consider them regulatory intermediaries without a direct impact on their production [10,47,48].

Cultural barriers: Deeply rooted beliefs about traditional production methods generate resistance to adopting new practices. Specifically, in communities with a strong campesino identity, technological extension is met with skepticism [49–52].

These results align with rural sociology studies that have documented mental models' influence on agricultural technology adoption [53]. The resistance to change observed in some producers is comparable to findings in other regions where cultural barriers limit the effectiveness of extension programs [38]. In this regard, the importance of designing extension strategies based on participatory approaches that consider local values and knowledge is emphasized [27,39,42,54].

II. Constraints and facilitators of technological adoption: a PESTEL perspective. We conducted an inquiry using ethnographic approaches to categorize constraints and facilitators into political, economic, social, technological, environmental, and legal (PESTEL) variables [22]. The objective was to understand producers' perspectives on their interaction with these variables, which may either limit or facilitate the implementation of sustainability-oriented practices, technologies, and innovations in oil palm cultivation in Colombia. Figure 7 presents the distribution of categorized variables according to the perceived relevance reported by producers.

The analysis revealed that economic factors are the most influential in technology adoption decisions, consistent with previous findings on the determinants of innovation in agricultural systems [9,38]. Producers reported that the investment required for adopting new technologies and uncertainty regarding profitability are the main barriers to implementing new practices, technologies, and innovations.

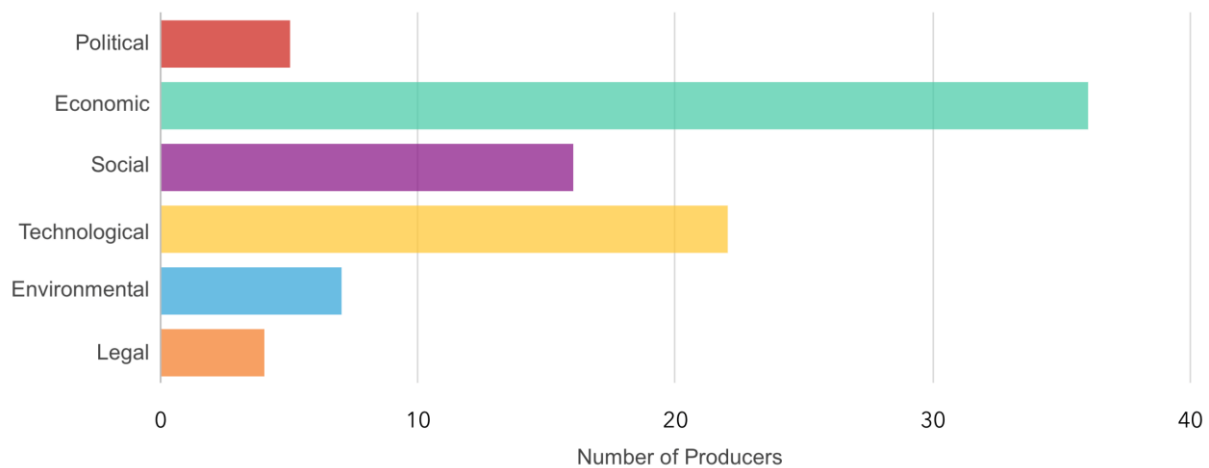


Figure 7. Constraints and facilitators of the adoption of practices, technologies, and innovations in oil palm cultivation in Colombia. This study employs a PESTEL perspective, drawing from ethnographic approaches with oil palm producers in Colombia.

Technological factors also emerged as key elements in decision making, suggesting that the availability of technologies adapted to local conditions and access to adequate infrastructure are critical determinants of the success of technology extension programs, aligning with the findings of Gatzweiler and Von Braun (2016) [41]. The lack of access to appropriate technologies and knowledge about their use was frequently mentioned as an obstacle, highlighting the need to strengthen training and technical support.

Regarding social factors, resistance to change and perceived risks associated with implementing new technologies were identified as significant constraints. This aligns with previous studies emphasizing the influence of social networks and organizational culture in agricultural technology adoption [10,35]. Specifically, producers with more significant experience in the sector exhibited higher resistance to modifying their traditional practices.

Environmental and legal factors were mentioned less frequently but remain relevant. While environmental regulations are essential for ensuring the sector's sustainability, some producers perceive them as bureaucratic barriers slowing innovation adoption. Similarly, legal certification frameworks and regulatory compliance generate additional costs that may discourage adopting new practices.

Finally, although political factors were the least mentioned, they should not be underestimated [38,43,55]. The stability of agribusiness policies and consistency in allocating incentives and subsidies can either facilitate or hinder the implementation of innovations in the oil palm sector. The variability of government policies has been documented as a challenge for long-term planning in agricultural technology adoption [30,38,43,55–57].

Based on a PESTEL perspective [22], this analysis demonstrates that adopting practices, technologies, and innovations in oil palm cultivation is a multifactorial and multidimensional phenomenon [12]. Economic and technological aspects dominate, while social, environmental, legal, and political dimensions are modulating factors. This is relevant for transforming mental models into an actual model, which will contribute to developing a dynamic hypothesis for the conceptual model proposed in this study on technology extension in agriculture in developing countries, particularly in Colombian oil palm cultivation.

3.1.4. Contribution of Reference Modes

The integrated analysis of the three reference modes enabled the construction of a conceptual framework for technology extension in the Colombian oil palm sector. Unlike previous models that have focused exclusively on technical/economic factors or considered the external factors of adoption as separate from extension systems, our research incorpo-

rated a multifactorial, multidimensional, and sociocultural understanding as key elements in technological adoption and legal and regulatory compliance.

The results have provided evidence of the need to design differentiated technology extension strategies according to producer profiles. Combining extension programs with participatory approaches can lead to higher adoption rates and long-term sustainability. These findings open new lines of research into the impact of interventions adapted to specific sociocultural contexts and their effectiveness in transforming production systems.

The integrated analysis of the reference modes revealed consistent patterns across the three data sources. Producers with low SI scores were also those with lower educational attainment and limited access to credit, as indicated in the socioeconomic characterization. These same producers emphasized economic barriers and uncertainty about returns as major obstacles to technology adoption in the ethnographic interviews. This mutual corroboration reinforces the reliability of the findings and highlights the value of using multiple converging sources to identify priority areas for targeted extension strategies.

3.2. Conceptualization of the Model and Analysis of Causal Relationships

The conceptual model developed in this study provides an analytical structure for understanding the dynamics of technology extension in oil palm cultivation in Colombia. This conceptualization explains how interactions between different system actors generate specific patterns of technological adoption and regulatory compliance. The integration of qualitative and quantitative approaches through the analysis of reference modes enabled the mapping of causal relationships and feedback loops within the system, facilitating the identification of key factors influencing its stability and evolution.

3.2.1. Identification of Actors and Their Interactions

Based on the ethnographic approaches described in Section 3.1.3, 15 key actors and their roles within the system were identified from the producers' perspective (Figure 8).

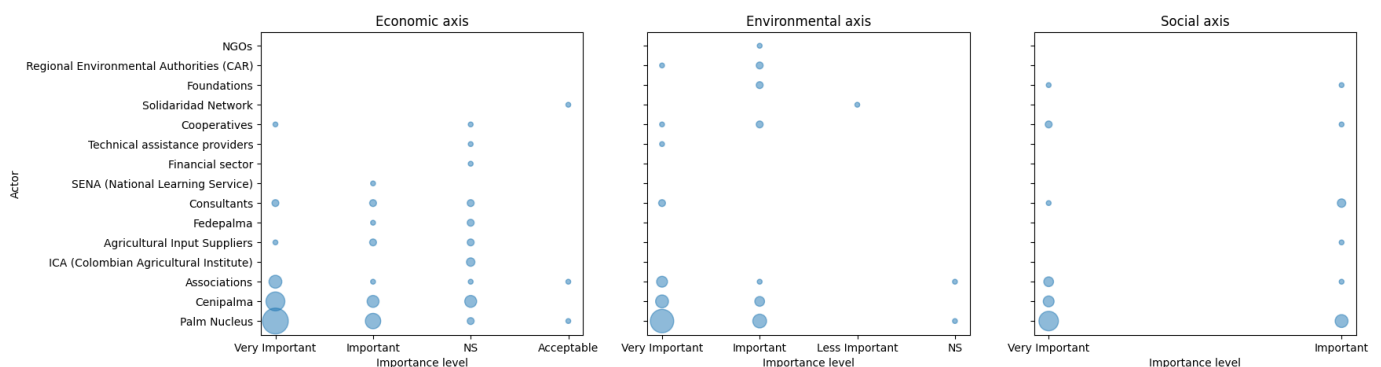


Figure 8. Count of actors identified by the sustainability axis and their level of importance. The diameter of the blue circles is directly proportional to the frequency with which each actor was mentioned during the ethnographic interviews. Larger circles indicate higher perceived relevance.

This figure was developed based on the frequency with which each actor was mentioned during the ethnographic interviews. The size of each circle reflects the number of times that actors were cited as relevant to achieving outcomes along the economic, environmental, and social sustainability axes. The classification of actors into these three dimensions was based on the thematic coding of their described functions and contributions, supported by triangulation with their formal institutional mandates and roles.

Table 2 describes the functionality and purpose of each actor identified in this research, including oil palm producers and workers as key actors.

Table 2. Actors related to the boundary of the iterative extension adoption technology model, identified through ethnographic approaches with producers.

Actors	Functionality and Purpose *
Oil Palm Producers and Workers	Fundamental actors in the oil palm sector's production chain, responsible for the primary production of palm fruit. Their work focuses on improving profitability and production sustainability through adopting technologies, compliance with legal regulations, and implementing beneficial agricultural practices.
Oil Palm Mill (Palm Nucleus)	An operational unit integrating processing plants and fruit suppliers, oriented towards sustainable productivity through technical support, compliance with the sustainability index, and the adoption of technologies.
Cenipalma	The research center for the oil palm sector is dedicated to the development and implementation of technologies and innovations to address sectoral challenges, with a focus on effective knowledge and technology transfer.
Associations	Organizational groups designed to generate economies of scale, facilitate collective bargaining, and promote the adoption of sustainable practices among their members.
ICA (Colombian Agricultural Institute)	The phytosanitary regulatory body responsible for establishing and enforcing regulations to ensure agricultural health and productivity through the implementation of appropriate practices.
Agricultural Input Suppliers	Specialized businesses in commercializing agricultural inputs, combining product sales with technical support to facilitate technology adoption.
Fedepalma	The federation that brings together palm oil producers in Colombia, promoting industry representation and leading investment initiatives focused on sector sustainability.
Advisors/Consultants	Professionals specializing in agronomic technical assistance, focusing on recommending practices that optimize sector productivity and sustainability.
SENA (National Learning Service)	An educational institution that trains agricultural workers, providing workforce training to facilitate technology adoption in the sector.
Financial Sector	Banking institutions that, through specialized credit lines, finance technology adoption by producers and support the viability of productive projects.
Technical Assistance Providers	Entities focused on training and technical advisory services for producers, aiming to strengthen their competitiveness and expand their market reach.
Cooperatives	Collective organizations that promote economic, environmental, and social sustainability through training strategies, hiring, and technology adoption in the oil palm sector.
Solidaridad Network	A non-governmental organization that supports small producers by managing international resources, prioritizing sustainability, and rural development projects.
Foundations	Organizations linked to companies in the sector that promote labor formalization and social well-being improvement through housing and quality-of-life projects.
CAR (Regional Environmental Authorities)	An environmental regulatory entity responsible for issuing and enforcing regulations to ensure the sustainability of productive activities.
NGOs	Organizations dedicated to rural development and quality of life improvement through international resource management and sustainability promotion in agricultural communities.

* Interpretation of functionality and purpose based on interactions with key stakeholders derived from the analysis of reference modes.

The identification of actors represented in Figure 8 enabled the establishment of causal technical relationships that explained the system's dynamics in terms of each sustainability axis and supported the structure of the conceptual model presented later.

3.2.2. Causal Relationships and Model Structure

The information presented in Table 3 was derived through a structured qualitative analysis. Initially, interview transcripts were coded to identify references to actors, their stated goals, and the processes underlying their decision making. These coded segments were grouped by actor type and sustainability dimension. Subsequently, decision rules were characterized by mapping the relationships among inputs, decision variables, and outputs, based on how actors described their roles and actions. The coding process was validated by cross-referencing with institutional documentation and complementary data sources.

The causal relationships among actors were defined through an iterative approach based on three phases [2,3,15]. First, the role(s) of each actor were determined according to the methodology described by Olaya (2016) [20]. Next, the interests and goals of each actor in relation to technological adoption and regulatory compliance were identified. For example, while Cenipalma focuses on generating technology and innovation, palm oil mills prioritize technical assistance and sustainable productivity. Finally, decision-making rules were characterized by determining the inputs and outputs of decisions made by each actor,

which allowed for the identification of the principles governing their behavior within the system (Table 3).

Table 3. Relationship between actors, goals, decision rules, and identified variables.

Actor	Sustainability Axis	Role	Goals and Interests	Decision Rules	Variables
Oil Palm Producers and Workers	Economic, Environmental, Social	Executor, Beneficiary	Increased profitability and sustainable productivity	Technological adoption, legal and regulatory compliance, SI compliance level	Training, affiliations, regulatory compliance, technology implementation
Oil Palm Mill (Palm Nucleus)	Economic, Environmental, Social	Collaborator, Beneficiary	Increase sustainable productivity	Strategic and operational plans, technical assistance to suppliers	Formalization of plans, technical support, advisory, SI compliance, technological adoption
Cenipalma	Economic, Environmental, Social	Executor, Stakeholder	Generate practices, technologies, and innovations	Strategic and operational plans, training and capacity building, coordination of phytosanitary management, sustainable and inclusive oil palm cultivation	Financing and formalization of plans, demonstration plots, productivity improvement plans, regulatory compliance, technical support, training, agreements, data collection, certified technical assistants, phytosanitary management, improvement of phytosanitary status, organizational models, SI compliance
Associations	Economic, Environmental, Social	Collaborator, Beneficiary, Responsible	Economies of scale and technological adoption	Producer organization, management, and economies of scale	Producer associativity, technical support, needs identification, economies of scale
ICA (Colombian Agricultural Institute)	Economic	Stakeholder	Maintain an adequate phytosanitary status	Issuance and enforcement of phytosanitary regulations	Regulatory framework, inspection, monitoring, and control, agreements
Agricultural Input Suppliers	Economic	Collaborator, Stakeholder	Increase sales of agricultural products	Sale of products and services	Product marketing, technical support, product application
Fedepalma	Economic, Environmental, Social	Collaborator, Stakeholder	Industry representation and administration of the palm oil fund	Producer association, sector investment projects	Producer affiliation, palm oil promotion fund, sector investment projects, sector advocacy, project implementation
Advisors/Consultants	Economic, Environmental, Social	Collaborator, Responsible	Technical support and profit	Technical assistance, client acquisition	Advisory, technical recommendations, advisory marketing, technical support
SENA (National Learning Service)	Economic, Environmental, Social	Stakeholder, Responsible	Training and technical capacity building for producers	Training programs for labor skills	Training programs, capacity building, agreements
Financial Sector	Economic, Environmental, Social	Stakeholder	Credit allocation for producers	Implementation of financial instruments	Application of financial instruments, credit allocation
Technical Assistance Providers	Economic	Collaborator, Stakeholder	Technical support for producers	Support and technical recommendations	Producer support, technological adoption, SI compliance
Cooperatives	Economic, Environmental, Social	Collaborator, Stakeholder	Promote technological adoption through organization	Producer and worker organization, labor formalization	Producer grouping, advisory, technical support, work schemes, memberships, skilled labor
Solidaridad Network	Economic, Environmental, Social	Stakeholder	Support small-scale producers and rural development	Agreements	Project management, training, collaborative work, agreements
Foundations	Economic, Environmental, Social	Stakeholder	Regulatory compliance and social improvement	Rural development projects	Project implementation, regional improvement, social management
CAR (Regional Environmental Authorities)	Environmental	Stakeholder	Environmental regulatory compliance	Issuance and enforcement of environmental regulations	Environmental regulations, environmental issues
NGOs	Environmental, Social	Stakeholder	Rural development and quality of life	International resource acquisition, deployment for adoption and compliance	Application for funding calls, training, resource management, agreements

The actor analysis confirmed the significance of a systemic approach to technological extension, aligning with previous studies on technology adoption in agriculture [9,38,58]. In particular, findings indicate that palm producers and workers, while being the primary executors of innovation implementation, heavily depend on the actions of other stakeholders to overcome economic and technical barriers [1–6].

Palm oil mills and technical advisors play a key role in technology transfer, a function well documented in the literature on agricultural extension systems, highlighting the importance of technical support in reducing uncertainty and increasing the acceptance

of innovations [6,10,59–61]. However, the lack of differentiated strategies according to producer profiles may limit the effectiveness of these efforts [39,42].

Regulatory entities and intervening organizations, such as Cenipalma and CAR, ensure the sector's sustainability through regulatory compliance and applied research. However, regulatory overload may generate resistance among producers, as documented in other agricultural sectors [38,43,55,57]. This highlights the need to balance regulatory requirements and incentive strategies to promote the voluntary adoption of sustainable practices.

Moreover, the role of the financial sector and commercial agricultural suppliers is fundamental in facilitating technological adoption [28,62,63]. Without adequate access to financing, producers' ability to implement innovations is significantly reduced, a finding consistent with studies on agricultural innovation and rural credit systems [12,28,39,52].

3.2.3. Conceptual Systemic Model of Technological Extension in Agriculture in Developing Countries

The conceptual systemic model presented in Figure 9 illustrates the interactions among different stakeholders and key variables involved in technological extension in agricultural systems within emerging economies. This model, grounded in the application of Sterman's iterative model (2000) [15], enables a comprehensive visualization of the interconnection between social, economic, and environmental sustainability through complex causal relationships that directly influence technological adoption levels and regulatory compliance in the production sector.

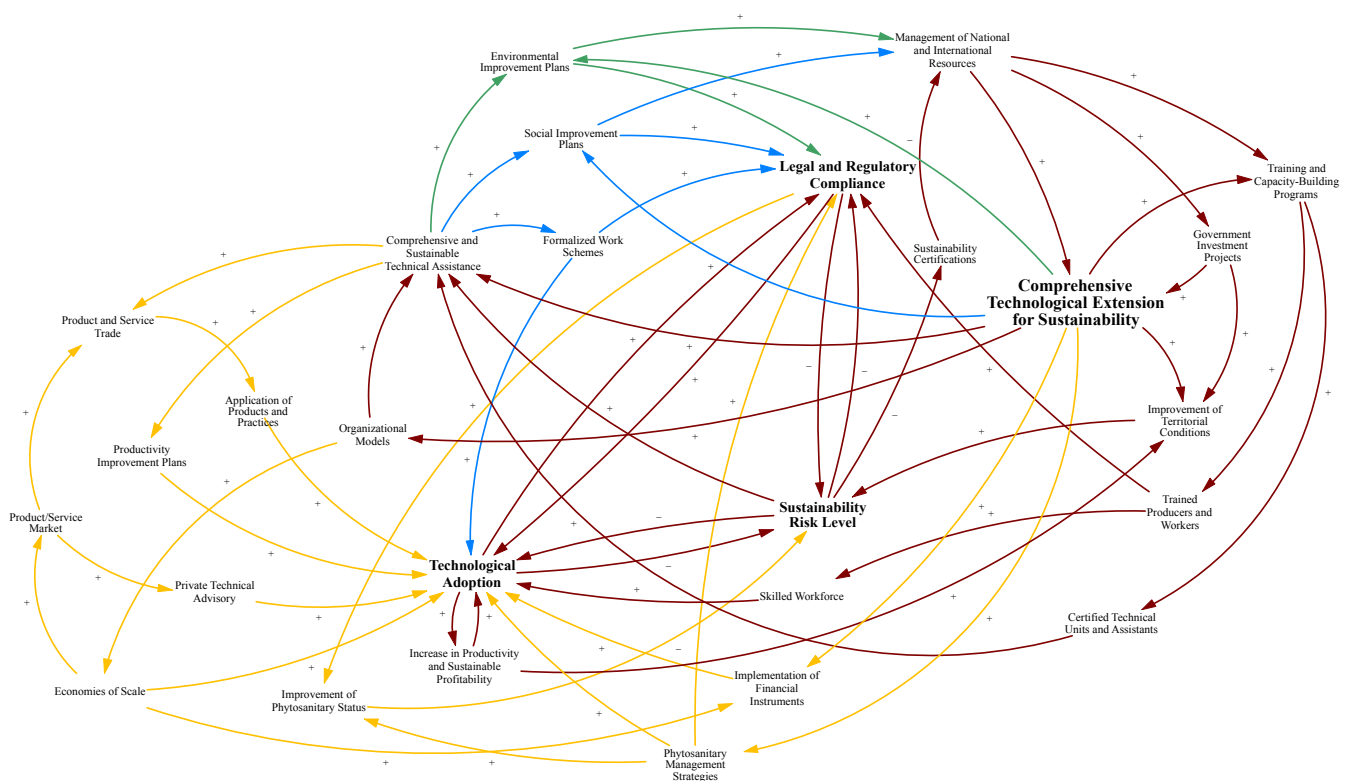


Figure 9. Conceptual systemic model of technological extension in agriculture in developing countries. Blue arrows: social axis; green arrows: environmental axis; yellow arrows: economic axis; brown arrows integrate the three dimensions of sustainability. Diagram generated using Vensim® PLE 10.2.2 (Ventana Systems, Inc., Harvard, MA, USA).

Figure 9 employs a color-coded system to represent causal relationships within the model: blue arrows indicate social interactions, green arrows represent environmental fac-

tors, yellow arrows depict economic relationships, and brown arrows highlight interactions that integrate all three dimensions of sustainability.

This research contributes significantly to understanding technology diffusion, extension, and adoption in agriculture in developing countries. Notably, the findings indicate that technological extension goes beyond mere technology transfer as it involves the following three highly interrelated and equally weighted key variables:

- Technological adoption.
- Legal and regulatory compliance.
- The level of risk associated with the sustainability index (SI).

3.2.4. Specific Conceptual Systemic Model for Technological Extension in Oil Palm Cultivation in Colombia

The conceptual model developed for technological extension in oil palm cultivation in Colombia provides a systemic perspective on the processes of technological adoption, regulatory compliance, and the level of risk associated with the sustainability index (SI) (Figure 10).

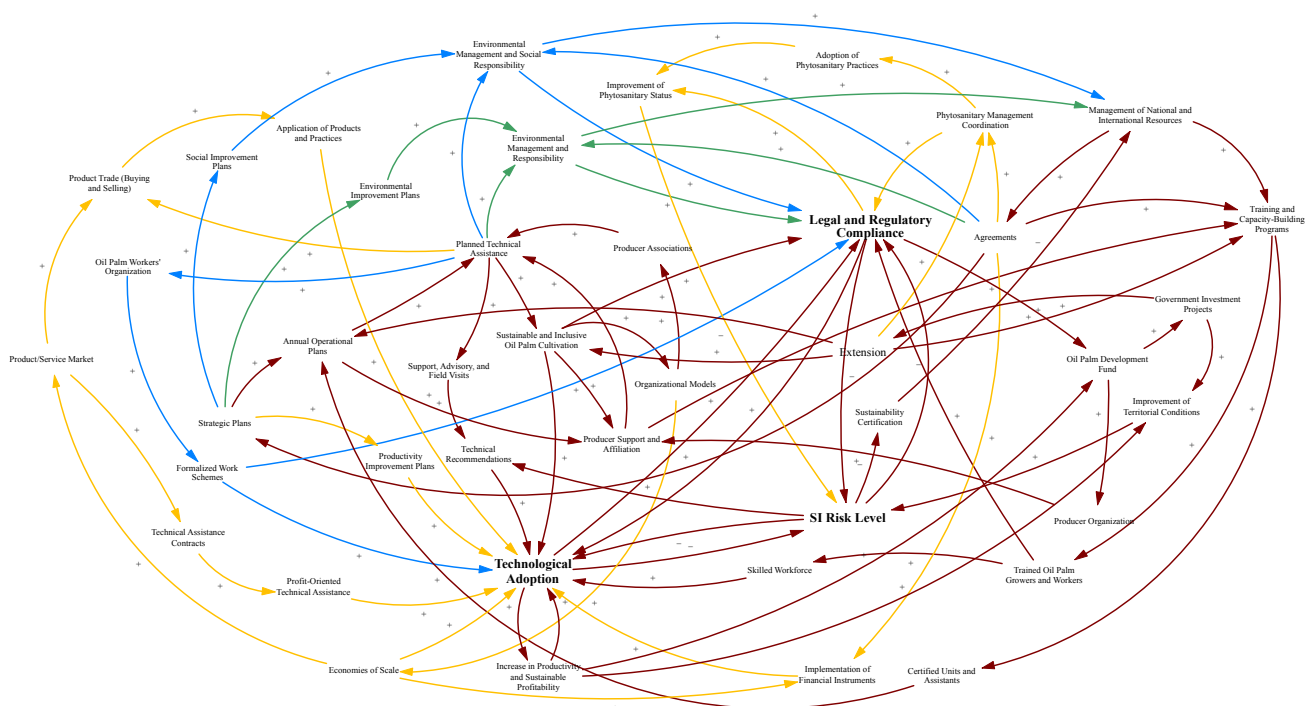
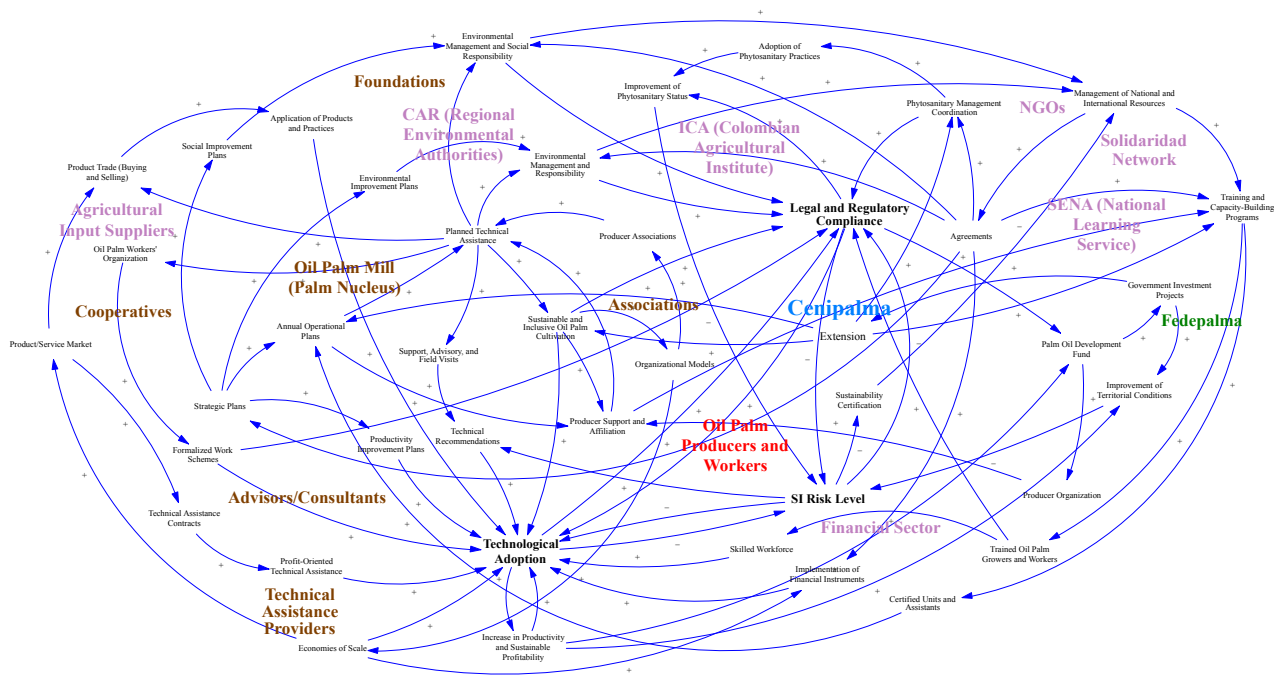


Figure 10. Specific conceptual systemic model for technological extension in oil palm cultivation in Colombia. Blue arrows: social axis; green arrows: environmental axis; yellow arrows: economic axis; brown arrows integrate the three dimensions of sustainability. Diagram generated using Vensim® PLE 10.2.2 (Ventana Systems, Inc., Harvard, MA, USA).

Unlike traditional approaches that address extension services from linear perspectives or consider contextual and environmental variables as external to an extension system [11,25,39,42], this model incorporates dynamic interactions among key stakeholders, integrating economic, social, and environmental variables, as shown in Figure 11, where each actor is positioned near its corresponding input variable in the system.



While all loops are embedded in the full model, selected representative examples are diagrammed and explained in detail later in this section to support interpretation and practical application.

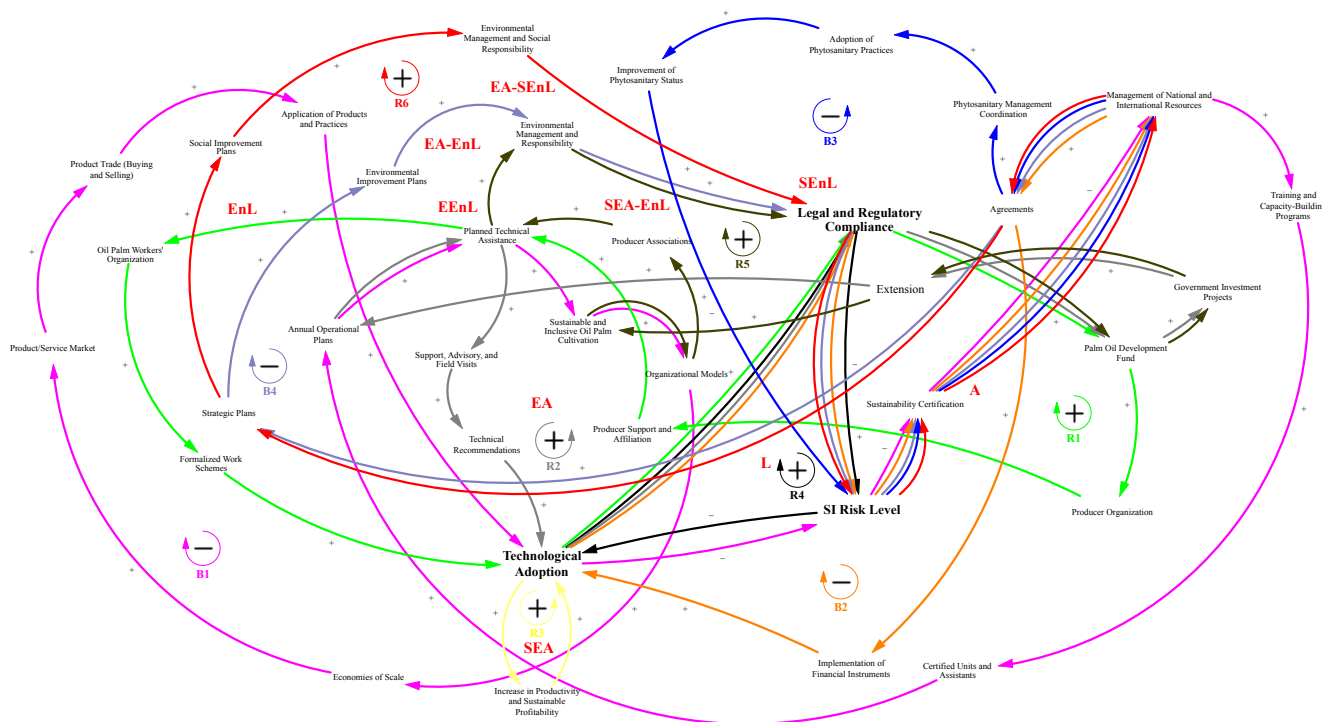


Figure 12. Feedback loops identified in the specific conceptual systemic model for technological extension in the oil palm sector in Colombia. The feedback loops, whether reinforcing (R) or balancing (B), are represented by sequences of arrows of the same color. Each arrow color corresponds to a specific causal loop to support visual differentiation. The red text overlays indicate producer typologies based on SI compliance level, as follows: A—advanced; L—lagging; EA—economically advanced; SEA—socioeconomically advanced; EnL—environmentally lagging; SENL—socio-environmentally lagging; EEnL—econo-environmentally lagging; EA-EnL—economically advanced/environmentally lagging; SEA-EnL—socioeconomically advanced/environmentally lagging; EA-SEnL—economically advanced/socio-environmentally lagging [14]. The validation was conducted through an iterative three-phase process, in which experts and sector stakeholders analyzed and refined the model by identifying key feedback loops and strategically positioning producer typologies. This approach enabled a deeper understanding of which variables exert the greatest impact on each group and which require priority intervention through extension strategies to promote technological adoption and regulatory compliance in the Colombian oil palm sector. Diagram generated using Vensim® PLE 10.2.2 (Ventana Systems, Inc., Harvard, MA, USA).

While Figure 12 presents a dense network of interactions, this level of complexity is essential to accurately represent the systemic dynamics of technological extension. The feedback loops and actor typologies are intentionally overlaid to allow for the visualization of interdependent relationships and intervention priorities within a realistic decision-making environment.

Previous studies have emphasized that technology adoption systems tend to fail when these factors are not adequately balanced, reinforcing the need to design intervention strategies based on dynamic models rather than static or linear approaches [1–4,6,15].

The overlay of producer typologies onto the systemic model was conducted to enhance the strategic utility of the model by linking specific feedback structures to real-world producer categories. This integration allows for targeted extension strategies based on

system dynamics rather than conventional demographic segmentation. The validation followed a structured three-phase participatory process.

To further enhance the interpretability of the feedback structures, Figure 13 presents a simplified depiction of two key loops extracted from the broader system: the reinforcing loop R1 and the balancing loop B1. Loop R1 illustrates how technological adoption fosters legal and regulatory compliance, which in turn increases collective contributions to the palm oil development fund. This strengthens producer organizations, enhances affiliation and technical assistance, and ultimately leads to greater adoption. In contrast, loop B1 shows how higher levels of sustainability certification enable access to national and international resources, which support capacity-building programs and increase the number of certified technical units. These units facilitate planned assistance and inclusive palm cultivation, promoting economies of scale and technology application. As technology adoption rises, the SI risk level declines, stabilizing the system. These examples illustrate how feedback dynamics shape leverage points for extension interventions.

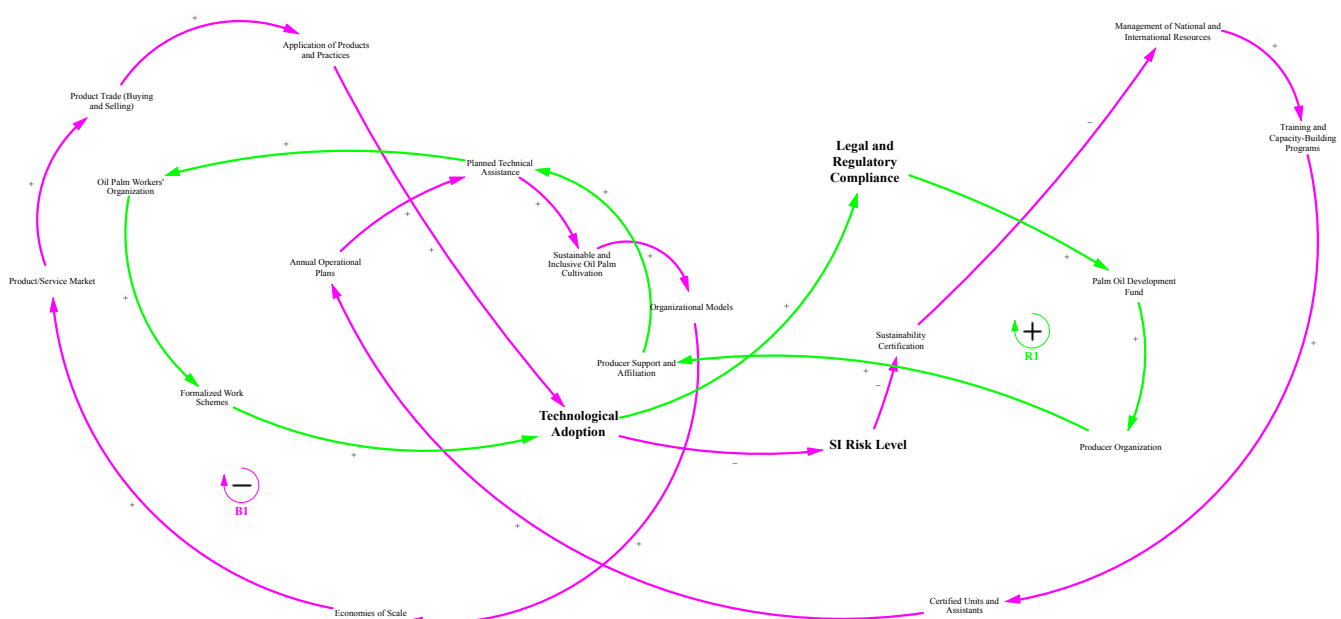


Figure 13. Representative feedback loops from the conceptual systemic model. This figure illustrates two representative feedback loops within the technological extension system: R1 (reinforcing) and B1 (balancing). The arrows represent the direction of causality, and the signs (+/−) indicate the nature of influence. Different arrow colors are used to distinguish the two loops for better visual interpretation. Diagram generated using Vensim® PLE 10.2.2 (Ventana Systems, Inc., Harvard, MA, USA).

The validation process for the overlay of producer typologies in Figure 12 was designed to ensure that their positioning accurately reflected real-world system dynamics. The process involved a total of 12 experts selected for their domain knowledge in economics, environmental sustainability, and social development within the oil palm sector. These experts represented diverse institutional affiliations, including Cenipalma, oil palm mills, environmental authorities, extension agents, and producer organizations. The iterative validation workshops combined individual analysis and group deliberation. Participants used both qualitative insights from field experience and quantitative reference data (SI profiles, socioeconomic indicators, adoption levels) to justify the placement of each producer type. Particular emphasis was placed on proximity to decision variables, interaction frequency, and the influence of key feedback loops in shaping the constraints and opportunities faced by each producer group.

The validation of these causal loops was carried out through an iterative three-phase process, involving experts in economic, environmental, and social aspects of the Colombian oil palm sector, along with key stakeholders engaged in the conceptual model. This process enabled the overlaying of producer typologies onto the model and the identification of priority feedback loops, ensuring that the conceptual structure accurately reflected the system's dynamics.

- First iteration: Each stakeholder independently analyzed the model, identifying the most relevant feedback loops based on their experience and overlaying the producer typologies onto the conceptual model. This phase provided an initial understanding of the relationship between typologies and system variables.
- Second iteration: A collective discussion was conducted among stakeholders and sectoral experts. During this phase, initial overlays were compared, inconsistencies were identified, and the analysis was enriched with different perspectives, leading to a more precise adjustment of the conceptual model.
- Third iteration: A final deliberation was conducted, during which stakeholders, through consensus and adjustments, established the definitive placement of each producer typology within the conceptual model. The proximity of each typology to specific variables helped identify the most influential factors and the areas requiring rapid intervention through extension strategies. This exercise facilitated the prioritization of actions aimed at moving producers from certain typologies toward higher levels of SI compliance.

As shown in Figure 12, the model enabled the overlay of producer typologies [14] based on their level of technology adoption and regulatory compliance. For instance, the “lagging” typology was positioned near variables related to economic risk and regulatory pressure, while “economically advanced” groups were closer to variables such as access to financing and affiliation strategies. These placements reflect the predominant feedback loops shaping each typology's behavior within the system. This segmentation supports differentiated extension strategies that go beyond demographic criteria (e.g., farm size or producer age) and are instead rooted in dynamic system interactions.

3.2.6. Practical Implications

The developed model has the potential to guide the formulation of public policies and private strategies for technological extension in oil palm cultivation and other agricultural sectors. Identifying causal loops and overlaying actors will enable the design of targeted interventions to reduce the SI risk level through incentive alignment, optimization of technical support, and adaptation to changing environments. Moreover, the model's ability to integrate political, economic, social, technological, environmental, and legal dimensions facilitates the implementation of more effective extension strategies, ensuring that technology adoption is not only financially viable but also aligned with the sector's sustainability objectives.

In this context, the system dynamics-based approach represents a key theoretical, methodological, and practical contribution to technological extension in agriculture in developing countries, providing an analytical framework to assess the impact of various policies and future interventions. This approach allows for strategic adjustments based on the specific needs of the productive sector where it is implemented.

Beyond its practical applications, the systemic model developed in this study contributes to the theoretical integration of key perspectives in agricultural innovation. First, it aligns with agricultural innovation systems (AIS) theory by explicitly mapping the roles, interactions, and influence of diverse stakeholders across regulatory, economic, and social dimensions [27,70,71]. Second, by identifying decision rules and perceived constraints

at the actor level, the model complements behavioral frameworks such as the technology acceptance model (TAM) and the unified theory of acceptance and use of technology (UTAUT), offering a systemic perspective on adoption determinants [72,73]. Lastly, the overlay of producer typologies and the emphasis on feedback structures highlight the relevance of social network theory, showing how structural position and connectivity influence technology diffusion [74,75]. These theoretical intersections enhance the model's explanatory power and applicability across varied contexts in agricultural development.

3.2.7. Limitations and Future Perspectives

Despite the advances achieved with this conceptual model, certain limitations should be considered in future research. First, the model is based on the information available at the time of this study, which may pose constraints on the representativeness of certain factors or the updating of key system variables. Additionally, although validation was conducted through an iterative process with sector stakeholders, broader empirical validation is still required through dynamic simulations and scenario analyses. These will allow for an assessment of the model's robustness under changing conditions in the productive environment.

Another limitation is the need to incorporate longitudinal data on reference modes, which would enable the evaluation of SI typologies' evolution and producers' socioeconomic characterization over time, along with their transition toward higher sustainability levels. Similarly, integrating quantitative and qualitative data through advanced modeling techniques such as the iterative steps 3, 4, and 5 of Sterman's model (2000) [15], actor network analysis, or artificial intelligence applied to agricultural decision making represents an opportunity to strengthen the model and expand its applicability to other productive sectors.

To address these limitations, future research will focus on constructing system dynamics-based simulation models that project the impact of different intervention strategies in the short, medium, and long term. Additionally, model validation through specific case studies in different oil palm-growing regions will allow for the refinement of critical variables and the incorporation of new influencing factors. Furthermore, developing digital tools for model implementation will facilitate its adoption by decision makers in the agricultural sector, maximizing its utility as a strategic analysis and planning platform for technological extension in dynamic environments.

4. Conclusions

This study provides a significant advancement in understanding technological extension in oil palm cultivation in Colombia, integrating multiple factors within a systemic approach based on system dynamics methodology. The findings indicate that technological adoption, regulatory compliance, and SI risk level cannot be analyzed in isolation; instead, they are highly correlated, acting as reinforcing and balancing variables contributing to system sustainability. Unlike previous studies that examine technology adoption through linear, unidirectional, or bidirectional perspectives influenced by external factors, our model captures the complexity of technological extension in agriculture in developing countries. It incorporates causal relationships and feedback loops, which explain how stakeholders' decisions impact the system's sustainability. This study overcomes the traditional paradigm that views extension as mere knowledge transfer, emphasizing its multidimensional and multifactorial nature and its key role in shaping productive systems. These findings significantly contribute to technological extension's evolution in the Colombian palm sector.

The findings of this research have direct implications for the formulation of public policies aimed at strengthening the sustainability of the Colombian oil palm sector. The developed model enables the design of differentiated technological extension strategies based on producer typologies, facilitating more effective interventions by addressing specific barriers and prioritizing training and financing programs for those with lower levels of technological adoption. Additionally, it provides a framework for assessing the impact of regulations on the adoption of sustainable technologies, allowing for the design of financial and tax incentives that promote the transition toward more sustainable production models, along with the optimization of certification policies and environmental monitoring. Furthermore, integrating this model into investment planning for rural infrastructure and market access helps identify strategic leverage points where government intervention can generate a greater impact. This approach ensures that financing programs consider the interactions between technical assistance, technology availability, and regulatory compliance. From an environmental perspective, the model facilitates the development of adaptation and climate change mitigation strategies through extension programs focused on agroecological practices, emission reduction, and the efficient use of natural resources.

This study demonstrates that technological extension in oil palm cultivation involves an interconnected network of actors with distinct roles whose behavior influences the effectiveness of technology adoption. By mapping actors and their interactions, this study identifies 16 key stakeholders, including producers, palm oil mills, technical advisors, technical assistance providers, environmental regulators, financial entities, and trade organizations. The overlapping of these actors in the specific model reveals that technology adoption depends on its availability and the actors' capacity to interact effectively within the system. Aligning their interests and strategies is crucial to overcoming economic, social, and regulatory barriers limiting the adoption of practices, technologies, and innovations, reinforcing the need for public and private policies promoting stakeholder synergies.

The identification of 10 feedback loops (6 reinforcing and 4 balancing) allows for an in-depth understanding of the dynamics shaping the evolution of the oil palm sector. The reinforcing loops, linked to technological adoption and regulatory compliance, explain how decision-making variables generate accumulative effects that strengthen the system. Additionally, SI risk-level feedback loops function as key balancing variables within the system. These findings highlight the importance of designing intervention strategies based on dynamic models rather than static or linear approaches, ensuring that technological extension policies remain effective in the long term. For instance, superimposing producer typologies within the systemic model enables the design of differentiated intervention strategies, ensuring that each group receives tailored support to enhance sustainability levels and reduce SI risk levels.

Building on this work, future research could advance the formulation of simulation models, the validation of results, and the design of intervention policies to enhance the sustainability of oil palm cultivation. This aligns with steps 3, 4, and 5 of the iterative model proposed by Sterman (2000) [15], allowing for evaluating and adjusting strategies based on the sector's evolving needs. This process lays the foundation for replicating the model in other agro-industrial systems with similar technology adoption structures.

This study marks a theoretical and methodological advance in the field of agricultural extension by introducing the first systems-based model that integrates technological adoption, regulatory compliance, and SI risk as interrelated core variables. The model's structure, grounded in feedback loops and actor-based typologies, provides a novel analytical lens that moves beyond traditional linear approaches. Its empirical foundation and system dynamics framework allow for an actionable segmentation of producers and a more precise understanding of the drivers and constraints of sustainability. As such, the

study contributes not only to extension theory and agricultural systems research but also offers a practical decision-support tool for policymakers and extension agents designing targeted interventions in dynamic rural environments.

Author Contributions: Conceptualization, J.F.B.-E., J.C.C. and L.H.R.; methodology, J.F.B.-E. and J.D.R.-O.; software, J.F.B.-E. and J.D.R.-O.; validation, J.C.C., L.H.R. and A.P.C.; formal analysis, J.F.B.-E.; investigation, J.F.B.-E., J.C.C., L.H.R. and A.P.C.; resources, J.F.B.-E.; data curation, J.F.B.-E. and J.D.R.-O.; writing—original draft preparation, J.F.B.-E.; writing—review and editing, J.F.B.-E., J.D.R.-O., J.C.C., L.H.R. and A.P.C.; visualization, J.F.B.-E.; supervision, J.C.C., L.H.R. and A.P.C.; project administration, J.F.B.-E.; funding acquisition, J.F.B.-E. and A.P.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Colombian Oil Palm Research Center Corporation (Cenipalma) with the support of the Colombian Fondo de Fomento Palmero (FFP). The project was financed through the FFP's investment program in technological extension for the oil palm sector, under funding codes EXT-2023 and EXT-2024.

Institutional Review Board Statement: This study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Research Committee of the Colombian Oil Palm Research Center—Cenipalma (protocol code 2024019000475H, approved on 8 May 2024). The ethical review was carried out under Cenipalma's internal research governance procedures in compliance with national guidelines for non-clinical research involving human participants in Colombia.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study. All participants were oil palm producers who voluntarily agreed to participate in semi-structured interviews after being informed of the purpose of the research, their rights as participants, and the confidentiality of their responses.

Data Availability Statement: The datasets presented in this article are not readily available because they include sensitive information from oil palm producers in Colombia, which is in the custody of Cenipalma. Requests to access the datasets should be directed to the Colombian Oil Palm Research Center Corporation (Cenipalma).

Acknowledgments: The authors thank the Colombian Oil Palm Research Center Corporation (Cenipalma), the Colombian Fondo de Fomento Palmero, and the Los Andes University in Colombia, which made this research possible. The authors would like to thank researchers Jorge Alonso Beltrán Giraldo and Paloma Bernal Hernández for their contribution to the conceptualization of this research and researchers Brayan Mauricio Rodríguez and Victor Orlando Rincón Romero for their contribution to the methodological development. The authors extend special thanks to the following Cenipalma extension agents for their contribution to conducting interviews: Miguel Ángel Hernández; Laura Milena Poveda; Paola Zarate; José Ricardo Toca; Angela Lucero Neira; Vivian Juliet González; Leidy Constanza Montiel; Carlos Alberto Ospina; Luz Janeth Cifuentes; Carlos Enrique Escobar; Blanca Yazmin Penagos; Adolfo José Núñez; William Alberto Mejía; Claudia Patricia Mendoza; Anderson Parada. The authors express their gratitude to the social extension team, composed of Oscar Alfredo Sanabria, Emy Johanna Arias, Zulma Dayana Matias, and Viviana Angelica de Leon from Cenipalma, for their contribution to the socioeconomic characterization of producers.

Conflicts of Interest: Authors Julián F. Becerra-Encinales and Alexandre P. Cooman were employed by the company Colombian Oil Palm Research Center Corporation—Cenipalma. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviation is used in this manuscript:

SI Sustainability Index

Appendix A

The following is the form used in the ethnographic approaches.

Table A1. Semi-structured interview form for oil palm producers.

Oil Palm Zone _____
Informed Consent _____
Name _____ Date _____
Nucleus _____ Sub-Zone _____
My name is [interviewer's name], and I am a staff member at Cenipalma. As part of our work in extension, we are conducting a research project titled "Design of a Management System for Technological Extension in Oil Palm". I would like to invite you to participate in this project, which will help us understand your needs in terms of technology transfer. If you agree to participate, I will ask you to allow me to interview you. The interview will last approximately one hour, during which I will ask questions about your experience in oil palm cultivation. Your participation in this research does not involve any material or financial compensation, and you are free to withdraw at any time. Your opinions and contributions to this research will be used exclusively for this project and stored securely. If you authorize me, I will record and transcribe the interview. If you prefer, your name will not appear in the study or any publication.
If you agree to participate in this project, please answer YES or NO to each of the following questions:
Do you agree to participate voluntarily in this project, understanding that you will not receive any material or financial compensation and that you can withdraw at any time? Yes _ No _
Do you authorize the use of photographs in which you appear in the research work and publications derived from this study? Yes _ No _
Do you authorize the recording of the interview and the taking of notes during the session? Yes _ No _
Do you request that your name not be revealed, and that your opinions, if cited, remain anonymous? Yes _ No _
Do you authorize your name to appear in the study or resulting publications to acknowledge your participation in this research or when your opinions are cited?
Yes _ No _
[Space for producer's signature] [Space for interviewer's signature]
Producer's Signature Interviewer's Signature
1. Socioeconomic Characteristics
Age: _____
Number of dependents economically supported by the producer: _____
Producer's educational level:
Primary _____ Secondary _____ Technical _____ University _____ Other _____
Years as a farmer: _____ Years as an oil palm grower: _____
Land ownership: Owned _____ Leased _____ Other _____
Age of oil palm plantation: _____ Number of hectares of oil palm: _____
Other crops: _____ Hectares: _____
Proportion of income derived from oil palm (%): _____
2. Technical Relationships
2.1. Which organizations, companies, or individuals have supported you with technological practices, training, or technical assistance impacting your productivity in the last 3 years?

Table A1. *Cont.*

(Agronomic management, harvest, and farm/plantation organization. Examples: innovative machinery, work organization, drainage, fertilization, sampling, pest and disease management, harvesting, post-harvest, residue utilization)
Name(s) of actor(s): 1. _____ 2. _____ 3. _____ ... N. _____
Importance rating:
1 = Low importance; 2 = Acceptable; 3 = Important; 4 = Very important
2.2. Which organizations, companies, or individuals have supported you with technological practices, training, or technical assistance in environmental issues in the last 3 years?
(Environmentally friendly practices related to soil conservation, water management, conservation of natural habitats, deforestation, waste management)
Name(s) of actor(s): 1. _____ 2. _____ 3. _____ ... N. _____
Importance rating:
1 = Low importance; 2 = Acceptable; 3 = Important; 4 = Very important
2.3. Which organizations or companies have supported you with technological practices, training, or technical assistance in social issues in the last 3 years?
(Relationship with the community, labor conditions—fair wages, contracts, social security, child labor, risks—organization, foreign workers, ethics, human rights, respect for other territories)
Name(s) of actor(s): 1. _____ 2. _____ 3. _____ ... N. _____
Importance rating:
1 = Low importance; 2 = Acceptable; 3 = Important; 4 = Very important
3. Verification for Quantifying Open-Ended Interview Responses
3.1. Experience and Evolution
Question: What does oil palm cultivation represent for you?
Category / Preliminary Score (Measurable or Quantifiable Aspects):
Life Purpose _____
Purely Business _____
Other (Specify): _____
Observations: (Space for interviewer's notes)
3.2. Technologies, Practices, or Innovations with Economic Impact
Question: Have you noticed any positive, negative, or no economic impact as a result of adopting technologies or practices in your oil palm plantation?
(Mention technologies or practices and the changes in productivity observed)
Technology/Practice/Innovation Name: _____
Category/Preliminary Score (Measurable or Quantifiable Aspects):
Increase: _____
Decrease: _____
No change: _____
Observations: (Space for interviewer's notes)
3.3. Technologies, Practices, or Innovations with Environmental Impact
Question: Have you noticed any positive, negative, or no environmental impact resulting from adopting technologies or practices in your oil palm plantation?

Table A1. *Cont.*

(Mention technologies or practices and their positive, negative, or neutral environmental impacts)
Technology/Practice/Innovation Name: _____
Positive Impact: _____
Negative Impact: _____
No Impact: _____
Observations: (Space for interviewer's notes)
3.4. Technologies, Practices, or Innovations with Social Impact
Question: Have you noticed any effects on the local community and workers due to adopting technologies or practices in your oil palm plantation?
(Mention technologies or practices and their effects on the local community and workers)
Category/Preliminary Score (Measurable or Quantifiable Aspects):
Technology/Practice/Innovation Name: _____
Positive Effect: _____
Negative Effect: _____
No Effect: _____
Observations: (Space for interviewer's notes)
3.5. Influencing Factors
Question: What motivates your decision to adopt or not adopt technologies, practices, or innovations in your oil palm plantation?
Category/Preliminary Score (Measurable or Quantifiable Aspects):
Economic Factors: _____
Access to technology/practice: _____
Technical knowledge: _____
Belief in technology: _____
Labor availability: _____
Other: Yes ___ No ___ If yes, specify: _____
Observations: (Space for interviewer's notes)
3.6. Challenges and Barriers
Question: What challenges or barriers have you encountered in adopting technologies, practices, or innovations?
Category/Preliminary Score (Measurable or Quantifiable Aspects):
Political: _____
Economic: _____
Social: _____
Technological: _____
Environmental: _____
Legal: _____
Observations: (Space for interviewer's notes)
3.7. Knowledge of Sustainability Certifications in Oil Palm
Question: Which sustainability certifications in oil palm are you aware of?
Category/Preliminary Score (Measurable or Quantifiable Aspects):

Table A1. *Cont.*

RSPO: __
APS Colombia: __
Other: Yes __ No __ If yes, specify: _____ (Multiple choice)
Observations: (Space for interviewer's notes)
3.8. Experience with Certification Programs
Question: What has been your experience participating in sustainability certification programs or initiatives?
Category/Preliminary Score (Measurable or Quantifiable Aspects):
Positive: _____
Regular: _____
Neutral: _____
Negative: _____ (Single choice)
Observations: (Space for interviewer's notes)
4. Additional Comments
Before concluding, is there anything else you would like to add? For example, how do you perceive the return on investment for implementing sustainable technologies, practices, and innovations in your business?

Appendix B

Below is the sustainability index (SI) for oil palm cultivation:

Table A2. Sustainability index of oil palm cultivation in Colombia.

ECONOMIC AXIS
PRINCIPLE 1. Productive, competitive, and resilient agribusiness
PRINCIPLE 6. Value generation from biomass
ENVIRONMENTAL AXIS
PRINCIPLE 2. Proper and efficient use of soil, water, and energy
PRINCIPLE 3. Zero deforestation and no replacement of HCV and HCS areas
PRINCIPLE 4. Harmonious palm cultivation with its natural surroundings and biodiversity
PRINCIPLE 5. Prevention and mitigation of environmental pollution
SOCIAL AXIS
PRINCIPLE 7. Decent employment and workers' rights
PRINCIPLE 8. Responsible supply schemes and inclusive businesses
PRINCIPLE 9. Responsible relationship with communities and human rights
PRINCIPLE 10. Ethical, legal, and transparent behavior

References

1. Trenberth, K.E.; Dai, A.; Van Der Schrier, G.; Jones, P.D.; Barichivich, J.; Briffa, K.R.; Sheffield, J. Global warming and changes in drought. *Nat. Clim. Change* **2014**, *4*, 17–22. [[CrossRef](#)]
2. Trautler, H.; Dubois, M.; Heikes, K.; Petiard, V.; Zilberman, D. *Megatrends in Food and Agriculture: Technology, Water Use and Nutrition*; John Wiley & Sons Ltd.: Hoboken, NJ, USA, 2018.
3. Aprillya, M.R.; Suryani, E.; Dzulkarnain, A. System Dynamics Simulation Model to Increase Paddy Production for Food Security. *J. Inf. Syst. Eng. Bus. Intell.* **2019**, *5*, 67–75. [[CrossRef](#)]
4. Turner, B.L.; Menendez, H.M.; Gates, R.; Tedeschi, L.O.; Atzori, A.S. System dynamics modeling for agricultural and natural resource management issues: Review of some past cases and forecasting future roles. *Resources* **2016**, *5*, 40. [[CrossRef](#)]

5. Sukiyono, K.; Romdhon, M.M.; Mulyasari, G.; Yuliarso, M.Z.; Nabiu, M.; Trisusilo, A.; Reflis; Napitupulu, D.M.T.; Nugroho, Y.; Puspitasari, M.S.; et al. The Contribution of Oil Palm Smallholders Farms to the Implementation of the Sustainable Development Goals-Measurement Attempt. *Sustainability* **2022**, *14*, 6843. [CrossRef]
6. Adebisi, J.A.; Olabisi, L.S. Participatory Causal Loop Mapping of the Adoption of Organic Farming in Nigeria. *Environ. Manag.* **2022**, *69*, 410–428. [CrossRef]
7. Fisher, D.K.; Norvell, J.; Sonka, S.; Nelson, M.J. Understanding technology adoption through system dynamics modeling: Implications for agribusiness management. *Int. Food Agribus. Manag. Rev.* **2000**, *3*, 281–296. [CrossRef]
8. Sharifzadeh, M.S.; Abdollahzadeh, G.; Damalas, C.A. Farmers' behaviour in the use of integrated pest management (IPM) practices: Perspectives through the social practice theory. *Int. J. Pest Manag.* **2023**, *69*, 1–14. [CrossRef]
9. Feder, G.; Anderson, J.R.; Ganguly, S. *The Rise and Fall of Training and Visit Extension: An Asian Mini-Drama with an African Epilogue*; World Bank Publications: Washington, DC, USA, 2006; Volume 3928. Available online: https://books.google.com.co/books?hl=es&lr=&id=MPgd9YUaDW4C&oi=fnd&pg=PA2&ots=ntGuaSYH1C&sig=f0VVjS5EtUfaBvPEdQOYWHjxCsg&redir_esc=y#v=onepage&q&f=false (accessed on 10 May 2022).
10. Leeuwis, C. *Communication for Rural Innovation: Rethinking Agricultural Extension*; Blackwell Science: Oxford, UK, 2005.
11. Rogers, E. *Diffusion and Innovations*; Free Press: Washington, DC, USA, 2010.
12. Becerra-Encinales, J.F.; Bernal-Hernandez, P.; Beltrán-Giraldo, J.A.; Cooman, A.P.; Reyes, L.H.; Cruz, J.C. Agricultural Extension for Adopting Technological Practices in Developing Countries: A Scoping Review of Barriers and Dimensions. *Sustainability* **2024**, *16*, 3555. [CrossRef]
13. Cenipalma, Perfil Socioeconómico, Base Para la Asistencia Técnica Planificada. Available online: <https://geoportal.cenipalma.org/caracterizacion/?page=Estadisticas> (accessed on 7 March 2025).
14. Becerra-Encinales, J.F. Compliance Patterns in Adopting Sustainability Practices: A Cluster Analysis of Oil Palm Producers in Colombia. Manuscript under review, 2025.
15. Sterman, J. *Business Dynamics: Systems Thinking and Modeling for a Complex World*; Irwin/McGraw-Hill: Chicago, IL, USA, 2000.
16. Bell, S.; Morse, S. Sustainability indicators past and present: What next? *Sustainability* **2018**, *10*, 1688. [CrossRef]
17. Mahecha, X. *Sustainable Palm Oil of Colombia (APSColombia): Aiming to Position Our Origin*; Fedepalma, Ed.; Fedepalma: Bogotá, Colombia, 2024. Available online: https://crmfedepalma-my.sharepoint.com/:p:/r/personal/xmahecha_apscolombia_org/_layouts/15/Doc.aspx?sourcedoc=%7B5B05C776-300D-4082-89A5-3A2A9CAE3074%7D&file=190224%20APSColombia%20strategy.pptx&fromShare=true&action=edit&mobileredirect=true (accessed on 1 April 2024).
18. Fedepalma. *Oil Palm in Colombia*; Fedepalma: Bogotá, Colombia, 2024.
19. Fedepalma; SISPA; SIFF; DANE; DIAN. La Palma de Aceite en Colombia. Available online: <https://repositorio.fedepalma.org/bitstream/handle/123456789/142819/Colombia%202022.pdf?sequence=1&isAllowed=y> (accessed on 1 April 2024).
20. Olaya, C.; Gómez-Quintero, J. Conceptualization of Social Systems: Actors First. In Proceedings of the 34th International Conference of the System Dynamics Society, Delft, The Netherlands, 17–21 July 2016; Universidad de los Andes: Bogotá, Colombia, 2016; pp. 1–18.
21. Olaya, C.; Guzmán-Abello, L. Diagramas de Ciclos Causales. In *Notas Sistémicas*; Universidad de los Andes: Bogotá, Colombia, 2022; Volume 1, pp. 1–7.
22. Mihailova, M. The state of agriculture in Bulgaria-PESTLE analysis. *Bulg. J. Agric. Sci.* **2020**, *26*, 935–943.
23. Fedepalma. La Palma de Aceite en Colombia. Available online: <https://web.fedepalma.org/la-palma-de-aceite-en-colombia-departamentos> (accessed on 4 April 2022).
24. Colombian Oil Palm Research Centre Corporation, Cenipalma; Becerra-Encinales, J.F.; Rodríguez, B.; Eloína, M.-F.; Beltrán-Giraldo, J.A.; Cooman, A.P.; Bernal-Hernández, P.; Reyes, L.; Cruz, J.C. Definition of Technological Extension Strategies Based on Exploratory Analysis of the Sustainability Index Using Artificial Intelligence: The Case of Oil Palm Producers in Colombia. Available online: <https://repositorio.fedepalma.org/handle/123456789/142851#page=1> (accessed on 2 April 2024).
25. Rogers, E.; Singhal, A.; Quinlan, M. *Diffusion of Innovations in An Integrated Approach to Communication Theory and Research*, 2nd ed.; Routledge: New York, NY, USA, 2014; Volume 2. Available online: <https://www.taylorfrancis.com/chapters/edit/10.4324/9780203887011-36/diffusion-innovations-everett-rogers-arvind-singhal-margaret-quinlan> (accessed on 31 May 2022).
26. Yuksel, I. Rogers' Diffusion of Innovation Model in Action: Individual Innovativeness Profiles of Pre-service Teachers in Turkey/Rogersov model difuzije inovacije u praksi: Pojedinačni profili inovativnosti budućih nastavnika u Turskoj. *Croat. J. Educ.—Hrvat. Čas. Odgoj. Obraz.* **2015**, *17*, 507–534. [CrossRef]
27. Klerkx, L.; Van Mierlo, B.; Leeuwis, C. Evolution of systems approaches to agricultural innovation: Concepts, analysis and interventions. In *Farming Systems Research into the 21st Century: The New Dynamic*; Springer: Dordrecht, The Netherlands, 2012; pp. 457–483.
28. Aguilar-Gallegos, N.; Muñoz-Rodríguez, M.; Santoyo-Cortés, H.; Aguilar-Ávila, J.; Klerkx, L. Information networks that generate economic value: A study on clusters of adopters of new or improved technologies and practices among oil palm growers in Mexico. *Agric. Syst.* **2015**, *135*, 122–132. [CrossRef]

29. Birkhaeuser, D.; Evenson, R.E.; Feder, G. *The Economic Impact of Agricultural Extension: A Review*; Center Discussion Paper, No. 567; Yale University, Economic Growth Center: New Haven, CT, USA, 1989. Available online: <https://www.econstor.eu/bitstream/10419/160489/1/cdp567.pdf> (accessed on 10 May 2022).
30. Swinnen, J.; Kuijpers, R. Value chain innovations for technology transfer in developing and emerging economies: Conceptual issues, typology, and policy implications. *Food Policy* **2019**, *83*, 298–309. [\[CrossRef\]](#)
31. Norton, G.W.; Alwang, J. Changes in Agricultural Extension and Implications for Farmer Adoption of New Practices. *Appl. Econ. Perspect. Policy* **2020**, *42*, 8–20. [\[CrossRef\]](#)
32. Norton, R. Política de Desarrollo Agrícola Conceptos y Principios. FAO: Rome, Italy, 2004; Volume 2.
33. Takahashi, K.; Muraoka, R.; Otsuka, K. Technology adoption, impact, and extension in developing countries' agriculture: A review of the recent literature. *Agric. Econ.* **2020**, *51*, 31–45. [\[CrossRef\]](#)
34. Beni Houd, Y.; El Amrani, M. Social Network Analysis: A useful tool for studying Innovation diffusion processes. *Econ. Agro-Aliment.* **2022**, *24*, 1–59. [\[CrossRef\]](#)
35. Bernal-Hernandez, P.; Ramirez, M.; Mosquera-Montoya, M. Formal rules and its role in centralised-diffusion systems: A study of small-scale producers of oil palm in Colombia. *J. Rural. Stud.* **2021**, *83*, 215–225. [\[CrossRef\]](#)
36. Taheri, F.; D'Haese, M.; Fiems, D.; Azadi, H. The intentions of agricultural professionals towards diffusing wireless sensor networks: Application of technology acceptance model in Southwest Iran. *Technol. Forecast. Soc. Change* **2022**, *185*, 122075. [\[CrossRef\]](#)
37. Thuo, M.; Bell, A.A.; Bravo-Ureta, B.E.; Lachaud, M.A.; Okello, D.K.; Okoko, E.N.; Kidula, N.L.; Deom, C.M.; Puppala, N. Effects of social network factors on information acquisition and adoption of improved groundnut varieties: The case of Uganda and Kenya. *Agric. Hum. Values* **2014**, *31*, 339–353. [\[CrossRef\]](#)
38. Pannell, D.; Zilberman, D. Understanding Adoption of Innovations and Behavior Change to Improve Agricultural Policy. *Appl. Econ. Perspect. Policy* **2020**, *42*, 3–7. [\[CrossRef\]](#)
39. Wigboldus, S.; Klerkx, L.; Leeuwis, C.; Schut, M.; Muilerman, S.; Jochemsen, H. Systemic perspectives on scaling agricultural innovations. A review. *Agron. Sustain. Dev.* **2016**, *36*, 46. [\[CrossRef\]](#)
40. Ali Mengal, A.; Habib, S.; Baloch, F.M.; Siddiqui, A.A.; Balochistan, Q. An innovative approach of public and private extension services regarding diffusion and adoption of agricultural technology in Balochistan, Pakistan: A conceptual framework. *JMAS J. MacroTrends Appl. Sci.* **2016**, *4*, 2016.
41. Gatzweiler, F.W.; Von Braun, J. *Technological and Institutional Innovations for Marginalized Smallholders in Agricultural Development*; Springer: Bonn, Germany, 2016. [\[CrossRef\]](#)
42. Klerkx, L.; Hall, A.; Leeuwis, C. Strengthening agricultural innovation capacity: Are innovation brokers the answer? *Int. J. Agric. Resour. Gov. Ecol.* **2009**, *8*, 409. [\[CrossRef\]](#)
43. Teece, D.J. Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy. *Res. Policy* **1986**, *15*, 285–305. [\[CrossRef\]](#)
44. Wejnert, B. Integrating Models of Diffusion of Innovations: A Conceptual Framework. *Annu. Rev. Sociol.* **2002**, *28*, 297–326. [\[CrossRef\]](#)
45. Amoussouhou, R.; Arouna, A.; Bavorova, M.; Tsangari, H.; Banout, J. An extended Canvas business model: A tool for sustainable technology transfer and adoption. *Technol. Soc.* **2022**, *68*, 101901. [\[CrossRef\]](#)
46. Walisinghe, B.R.; Ratnasiri, S.; Rohde, N.; Guest, R. Does agricultural extension promote technology adoption in Sri Lanka. *Int. J. Soc. Econ.* **2017**, *44*, 2173–2186. [\[CrossRef\]](#)
47. Benor, D.; Harrison, J.Q. *Agricultural Extension: The Training and Visit System*; World Bank: Washington, DC, USA, 1977.
48. Gonzáles, H. Hacia una Nueva Concepción de la Extensión; 2000. Available online: <https://www.mag.go.cr/bibliotecavirtual/AV-1700.pdf> (accessed on 10 February 2025).
49. Gardezi, M.; Abuayyash, H.; Adler, P.R.; Alvez, J.P.; Anjum, R.; Badireddy, A.R.; Brugler, S.; Carcamo, P.; Clay, D.; Dadkhah, A.; et al. The role of living labs in cultivating inclusive and responsible innovation in precision agriculture. *Agric. Syst.* **2024**, *216*, 103908. [\[CrossRef\]](#)
50. Huergo, J. Desafíos de la extensión desde la perspectiva cultural. *Dialoguemos* **2004**, *8*, 11–15.
51. Martínez-Arteaga, D. *Determinantes de la Adopción de Tecnologías para el Manejo Eficiente del Agua por los Cultivadores de Palma de Aceite en la Zona Norte Colombiana*; Universidad Nacional de Colombia: Bogotá, Colombia, 2022.
52. Martínez-Arteaga, D.; Arias Arias, N.A.; Darghan, A.E.; Barrios, D. Identification of Influential Factors in the Adoption of Irrigation Technologies through Neural Network Analysis: A Case Study with Oil Palm Growers. *Agriculture* **2023**, *13*, 827. [\[CrossRef\]](#)
53. Jansen, J.; Steuten, C.D.; Renes, R.J.; Aarts, N.; Lam, T.J. Debunking the myth of the hard-to-reach farmer: Effective communication on udder health. *J. Dairy Sci.* **2010**, *93*, 1296–1306. [\[CrossRef\]](#) [\[PubMed\]](#)
54. Chambers, R.; Ghildyal, B.P. Agricultural research for resource-poor farmers: The farmer-first-and-last model. *Agric. Adm.* **1985**, *20*, 1–30. [\[CrossRef\]](#)

55. Hazell, P.; Poulton, C.; Wiggins, S.; Dorward, A. The Future of Small Farms: Trajectories and Policy Priorities. *World Dev.* **2010**, *38*, 1349–1361. [\[CrossRef\]](#)
56. Gorton, M.; Douarin, E.; Davidova, S.; Latruffe, L. Attitudes to agricultural policy and farming futures in the context of the 2003 CAP reform: A comparison of farmers in selected established and new Member States. *J. Rural. Stud.* **2008**, *24*, 322–336. [\[CrossRef\]](#)
57. Jadallah, A.E.O.; Hassan Abu Bakar, A.; Hsnah Mohd, J. The impact of the decentralization and Pluralism policy on agricultural extension services. *J. Agric. Technol.* **2011**, *7*, 895–901.
58. Anderson, J.R.; Feder, G. Chapter 44 Agricultural Extension. *Handb. Agric. Econ.* **2007**, *3*, 2343–2378. [\[CrossRef\]](#)
59. Anastasiadis, F.; Tsolakis, N.; Srai, J.S. Digital technologies towards resource efficiency in the agrifood sector: Key challenges in developing countries. *Sustainability* **2018**, *10*, 4850. [\[CrossRef\]](#)
60. Woittiez, L.S.; van Wijk, M.T.; Slingerland, M.; van Noordwijk, M.; Giller, K.E. Yield gaps in oil palm: A quantitative review of contributing factors. *Eur. J. Agron.* **2017**, *83*, 57–77. [\[CrossRef\]](#)
61. Parveez, G.K.A.; Hishamuddin, E.; Loh, S.K.; Ong-Abdullah, M.; Salleh, K.M.; Bidin, M.N.I.Z.; Sundram, S.; Hasan, Z.A.A.; Idris, Z. Oil palm economic performance in malaysia and R&D progress in 2019. *J. Oil Palm Res.* **2020**, *32*, 159–190. [\[CrossRef\]](#)
62. Bilal, M.; Brümmer, B.; Barkmann, J. A discussion on the outcomes of adopted agricultural technological products and specific sustainable development goals: Evidence from Pakistan. *Cogent Econ. Financ.* **2022**, *10*, 1–21. [\[CrossRef\]](#)
63. Nazar, H.; Ullah, S.; Nasir, S.; Bilal, M. Exploring the potential determinants to favour available entrepreneurial strategies among dairy farmers of southern Punjab in Pakistan. *J. Agric. Educ. Ext.* **2024**, *31*, 71–89. [\[CrossRef\]](#)
64. Abdulai, A. Information acquisition and the adoption of improved crop varieties. *Am. J. Agric. Econ.* **2023**, *105*, 1049–1062. [\[CrossRef\]](#)
65. Arslan, C.; Wollni, M.; Oduol, J.; Hughes, K. Who communicates the information matters for technology adoption. *World Dev.* **2022**, *158*, 106015. [\[CrossRef\]](#)
66. Bonjean, I. Heterogeneous incentives for innovation adoption: The price effect on segmented markets. *Food Policy* **2019**, *87*, 101741. [\[CrossRef\]](#)
67. Omar, J.A.E.; Bakar, A.H.A.; Jais, H.M.; Shalloof, F.M. Study of the Role of Agricultural Extension in the Dissemination of Sustainable Agricultural Development. *J. Agric. Technol.* **2012**, *8*, 1237–1245.
68. Janssen, E.; Swinnen, J. Technology adoption and value chains in developing countries: Evidence from dairy in India. *Food Policy* **2019**, *83*, 327–336. [\[CrossRef\]](#)
69. Meinzen-Dick, R.; Quisumbing, A.R.; Behrman, J.A. A System That Delivers: Integrating Gender into Agricultural Research, Development, and Extension. In *Gender in Agriculture*; Springer: Dordrecht, The Netherlands, 2014; pp. 373–391. [\[CrossRef\]](#)
70. Swanson, B.E.; Rajalahti, R.; The World Bank. Strengthening Agricultural Extension and Advisory Systems: Procedures for Assessing, Transforming, and Evaluating Extension Systems. 2010. Available online: <http://www.worldbank.org/rural> (accessed on 14 July 2024).
71. World Bank. *Enhancing Agricultural Innovation: How to Go Beyond the Strengthening of Research Systems*; The World Bank: Washington, DC, USA, 2006. [\[CrossRef\]](#)
72. Davis, F.D. Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Q.* **1989**, *13*, 319. [\[CrossRef\]](#)
73. Venkatesh, V.; Morris, M.G.; Davis, G.B.; Davis, F.D. User Acceptance of Information Technology: Toward a Unified View. *MIS Q.* **2003**, *27*, 425. [\[CrossRef\]](#)
74. Borgatti, S.P.; Mehra, A.; Brass, D.J.; Labianca, G. Network Analysis in the Social Sciences. *Science* **2009**, *323*, 892–895. [\[CrossRef\]](#)
75. Valente, T.W. Social network thresholds in the diffusion of innovations. *Soc. Netw.* **1996**, *18*, 69–89. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.